

ECSE 4560/ 6560

Modern Communication Systems

Lecture 21: Wireless Sensing Part 2: Localization

Ish Jain, Fall 2025



Announcements

- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **Localization**
- Projects
 - Group formation [MCS_Fall25_project.xlsx](#)
 - Project mid-term report due **Monday 11/10**
 - Final Project Presentation and Demos: Dec 9 and Dec 12
 - Project Report and peer evaluation: Finals exam week
- HW/Exams Dates
 - HW5 (MIMO Theory) due **Mon 11/10**. Late due (Wed 11/13)
 - HW6 (MIMO code) due **Fri 11/14**
 - HW7 (Sensing Theory) due **Fri 11/21** – covered in Exam 2
 - Exam 2: **Friday Nov 21** - Anyone needs makeup exam?

Extra Reading on Piazza

- Principles of Modern Wireless Communication Systems Theory and Practice; by Aditya K. Jagannatham, IIT Kanpur Book
 - https://piazza.com/class_profile/get_resource/mevvk9kirhg2fu/mgb9esiush64ae
 - Parts of Chapter 1,2,3,4,7
 - **MIMO Chapter 6**
- Sensing Readings
 - Xiong, Jie, and Kyle Jamieson. "ArrayTrack: A Fine-Grained indoor location system." *NSDI 2013*
 - 2024 SIGMOBILE Test-of-Time Award 

Today's Class

- MIMO Review (HW 6)
- Clock Synchronization in MIMO
- Intro-Wireless Sensing
 - Localization Techniques

Today's Class

- MIMO Review (HW 6)
- Clock Synchronization in MIMO
- **Intro-Wireless Sensing**
 - **Localization Techniques**

Wireless Localization / Positioning

The process of obtaining a human or object's location using wireless signals

Applications:

- Navigation: outdoors (GPS) and indoors (e.g., museum)
- Location based services: Tagging, Reminder, Ads
- Virtual Reality and Motion Capture
- Gestures, writing in the air
- Behavioral Analytics (Health, activities, etc.)
- Locating misplaced items (keys)
- Location based security
- Delivery drones

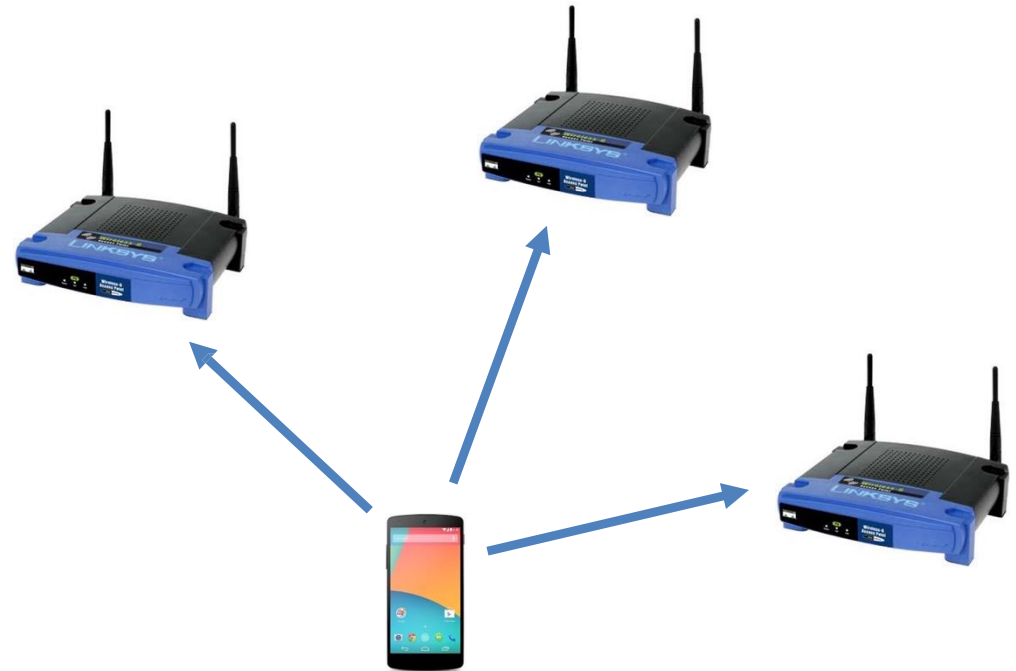


Wireless Localization Architecture.

- Device based: A device uses incoming signal from one or more “anchors” to determine its own location

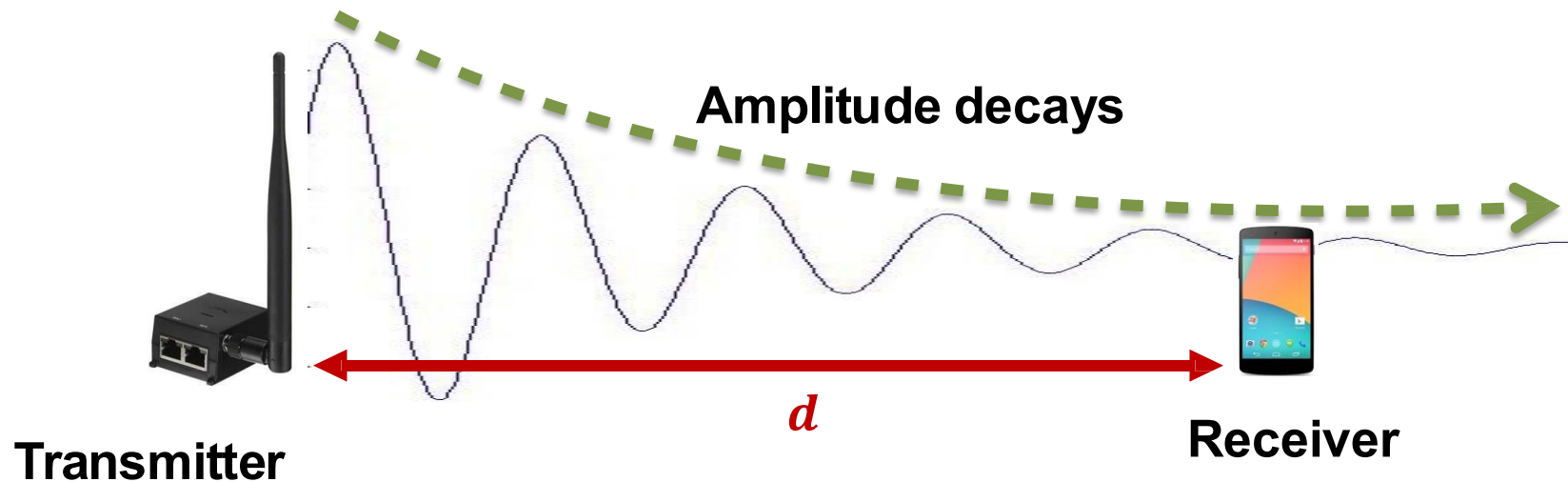


- Network based: Anchors (or Access points) use the signal coming from device to determine its location



Method 1: RSSI Based Localization

- Higher received power → Closer
- Lower received power → Farther

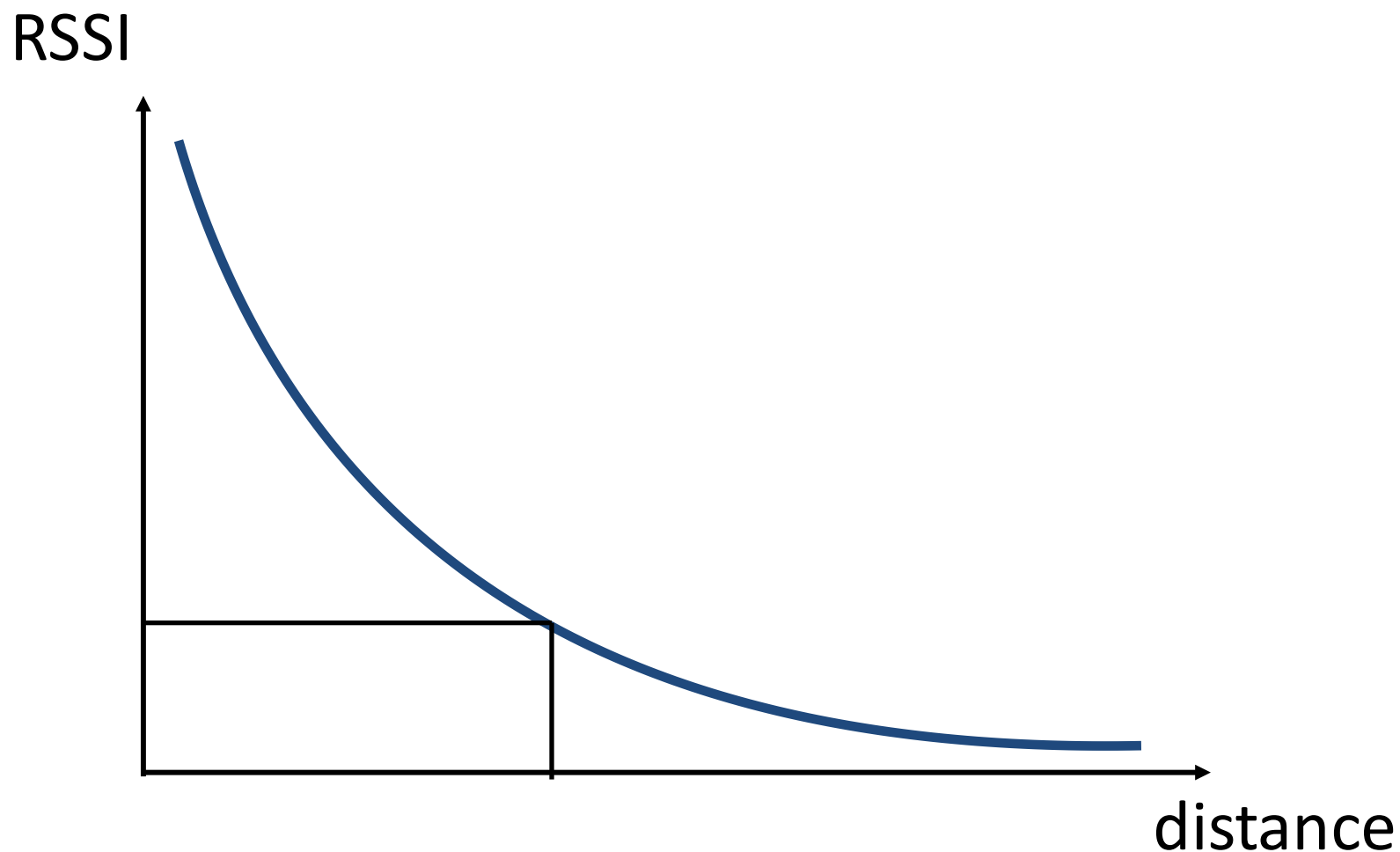


$$P_{Rx} = \frac{G_{Tx} G_{Rx} \lambda^2}{(4\pi d)^2} P_{Tx} \quad \Rightarrow \quad RSSI \propto \frac{1}{d^2}$$

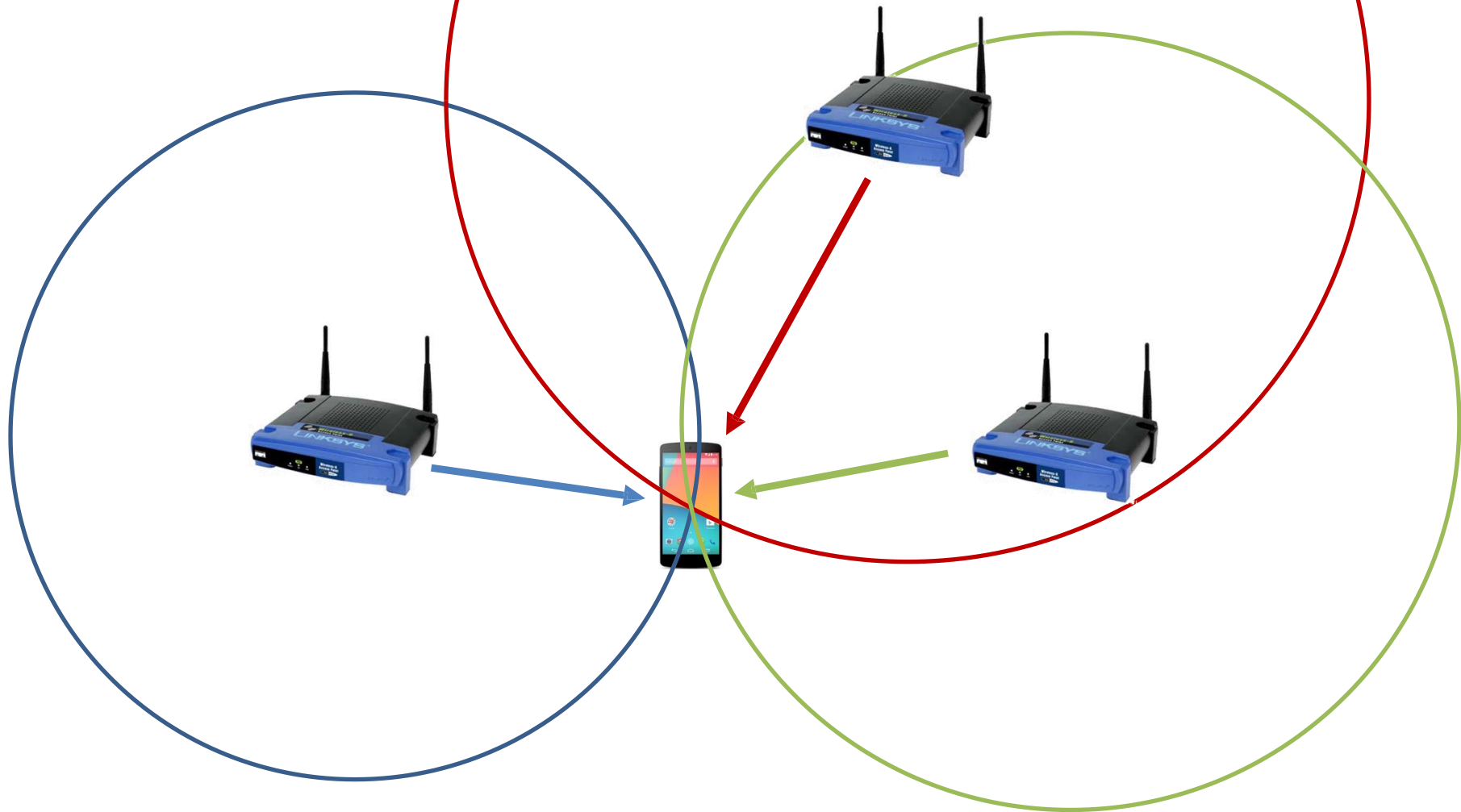
Use RSSI to estimate distance from APs!

Method 1: RSSI Based Localization

$$RSSI \propto \frac{1}{d^2}$$



Method 1: RSSI Based Localization

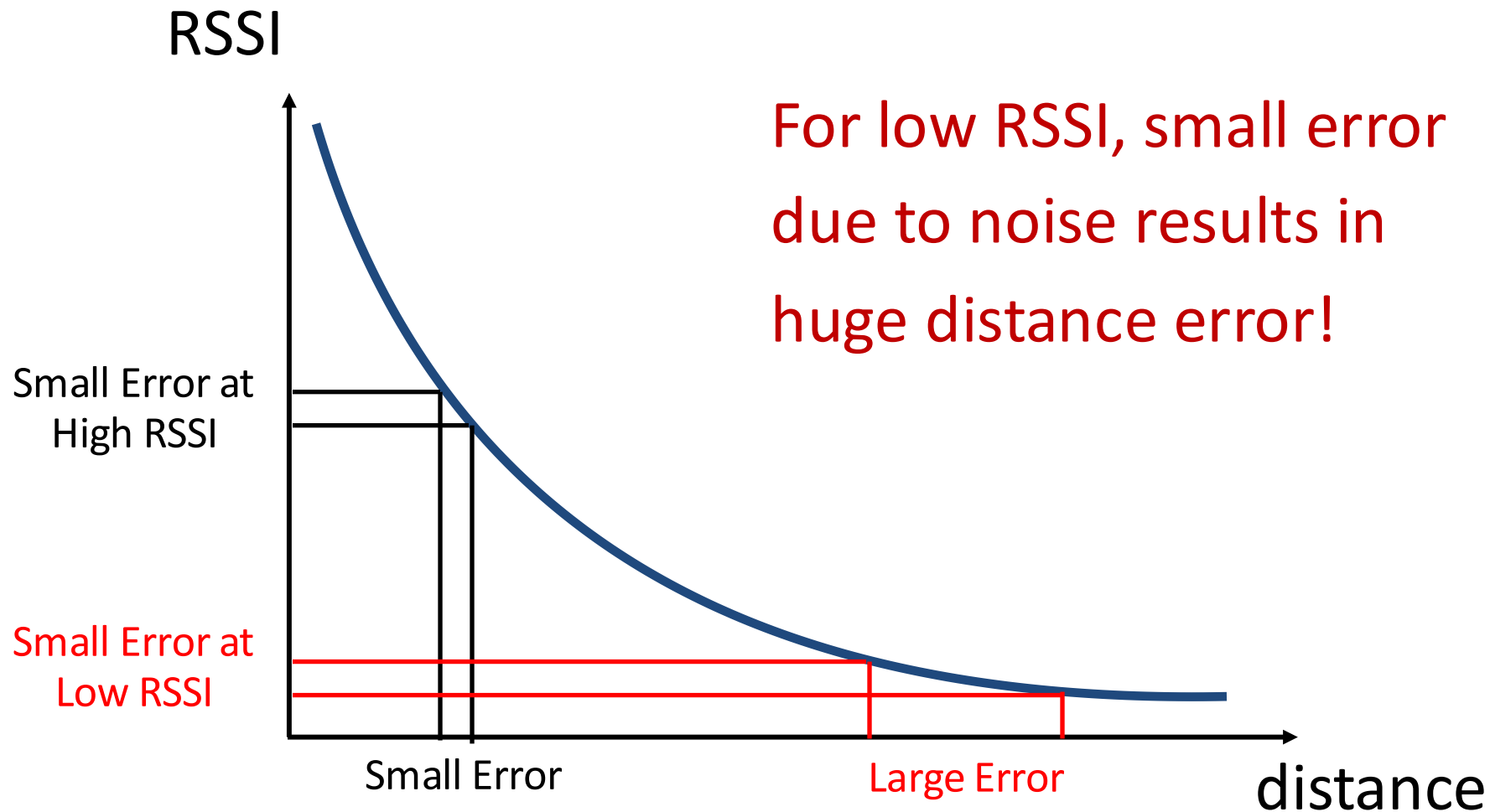


Trilateration

Method 1: RSSI Based Localization

Pros: Very simple, no hardware modifications

Cons: Highly inaccurate!



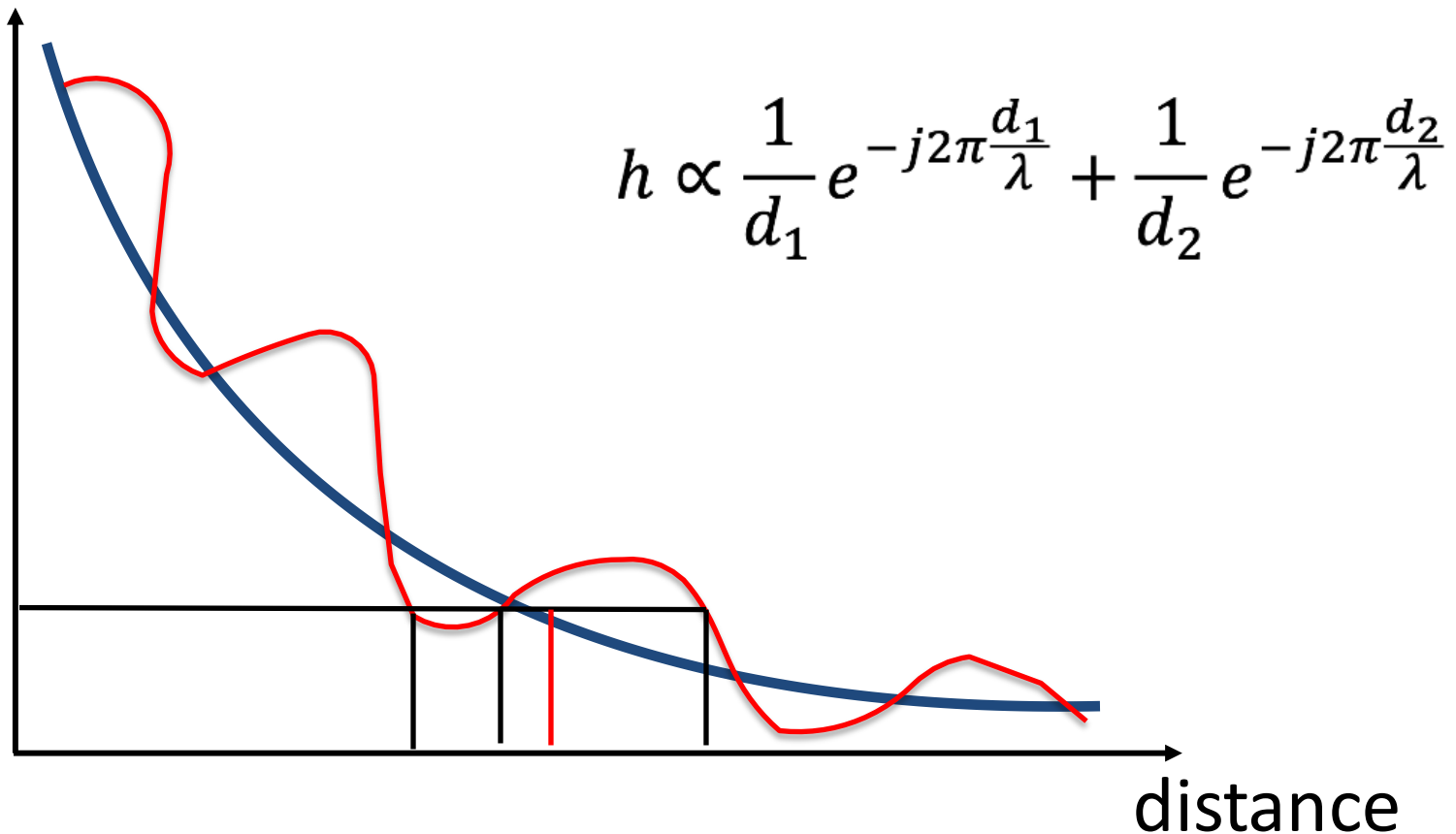
Method 1: RSSI Based Localization

Pros: Very simple, no hardware modifications!

Cons: Highly inaccurate!

Does not work with multipath!

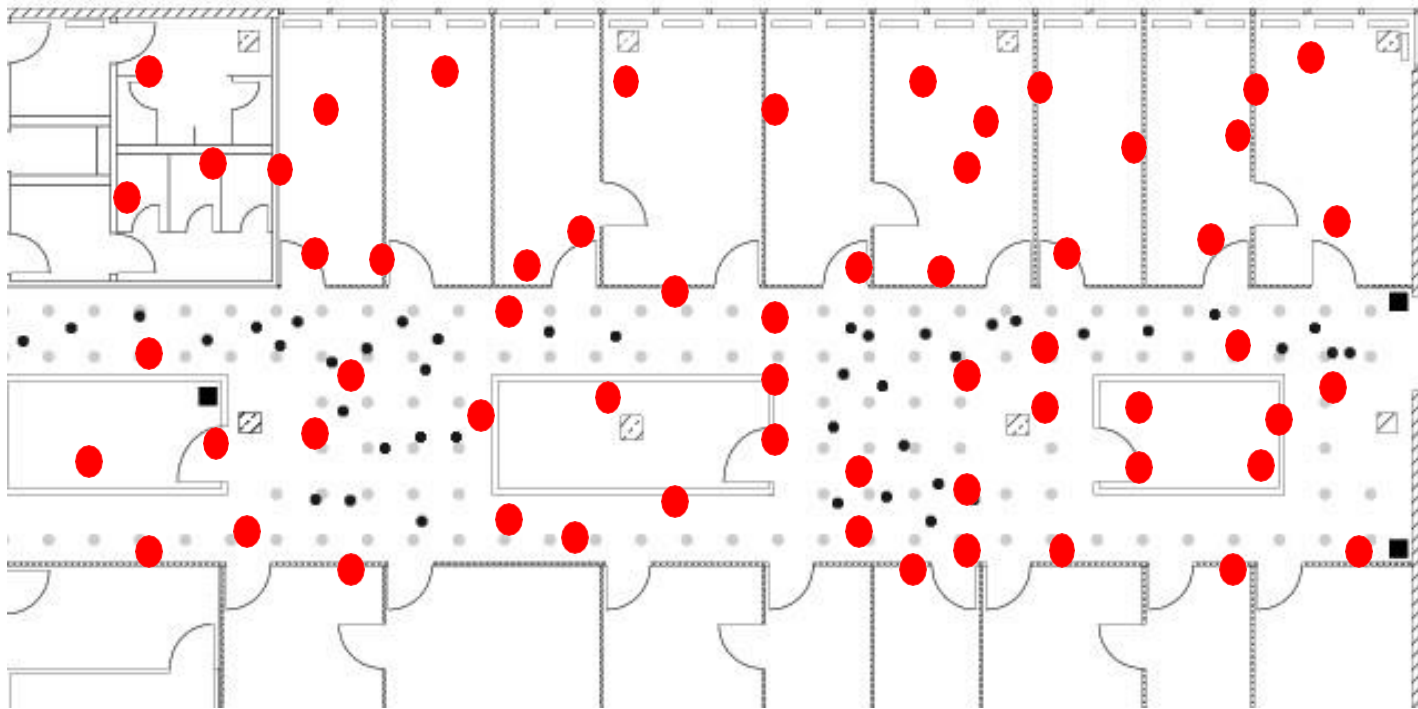
RSSI



Method 1: RSSI Based Localization

Solution: Fingerprinting

Measure and records RSSI fingerprints at each location (war-driving)

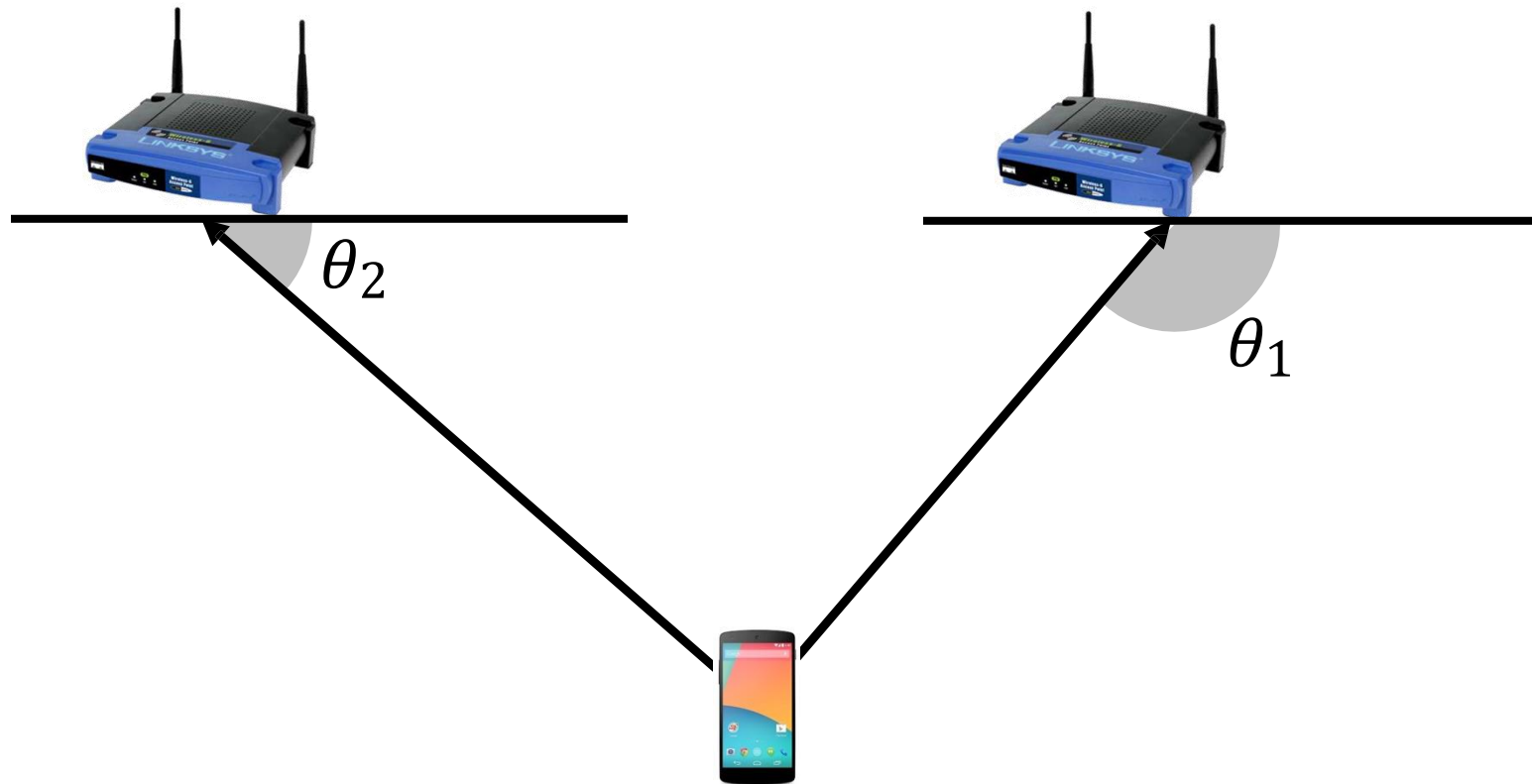


Pros: Works with multipath, No need to know AP locations!

Cons: Changes in environment/movement → change RSSI!
Continuous training is needed. Lots of effort!

Method 2: AoA Based Localization

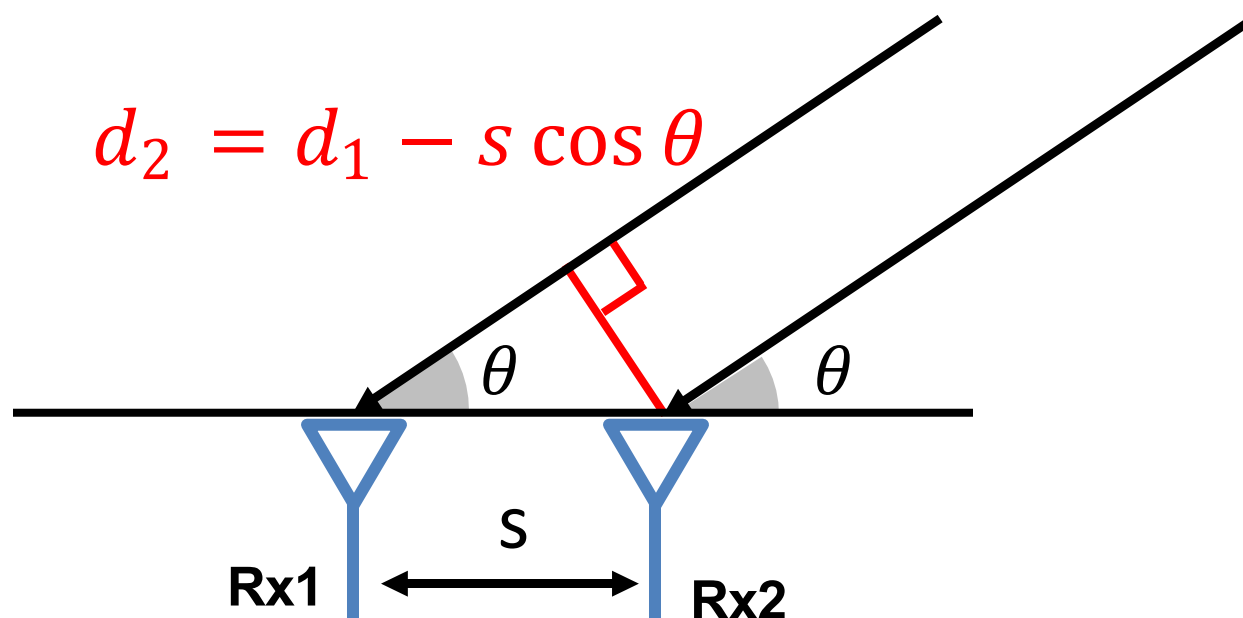
Measure Angle of Arrival (AoA) from device to each AP



Triangulation

Method 2: AoA Based Localization

Measure Angle of Arrival (AoA) from device to each AP

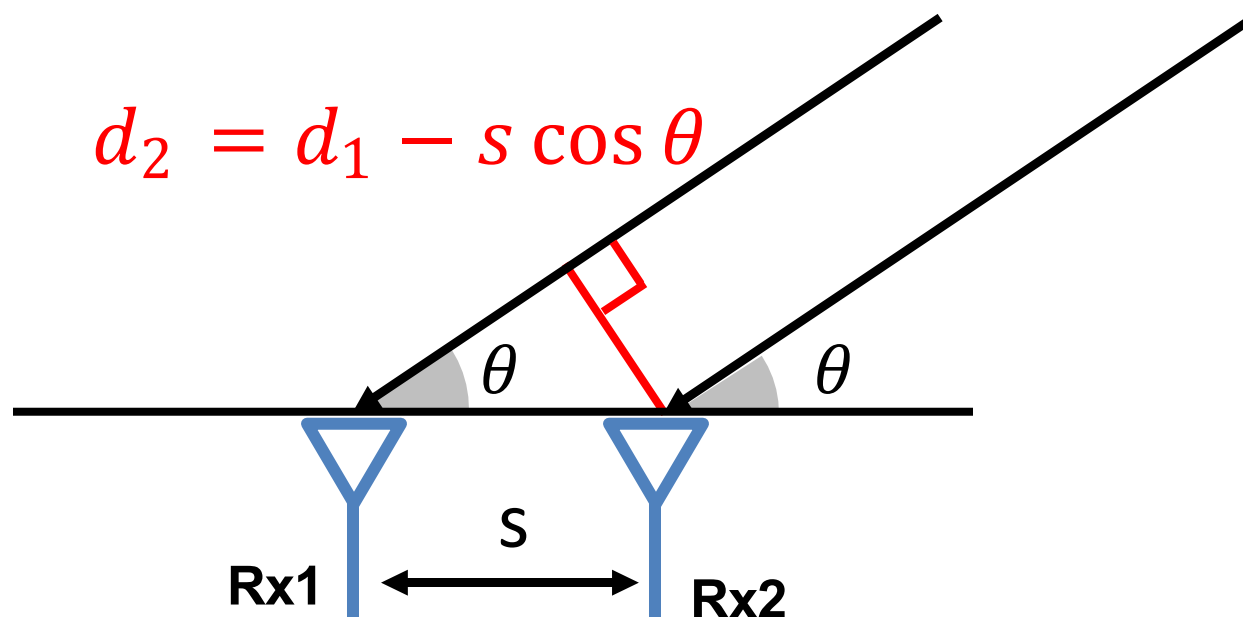


$$h_1 \propto e^{-j2\pi \frac{d_1}{\lambda}} \quad h_2 \propto e^{-j2\pi \frac{d_2}{\lambda}} = e^{-j2\pi \frac{d_1 - s \cos \theta}{\lambda}}$$

$$\Delta\Phi = \angle h_2 - \angle h_1 = 2\pi s \cos \theta / \lambda \bmod 2\pi$$

Method 2: AoA Based Localization

Measure Angle of Arrival (AoA) from device to each AP



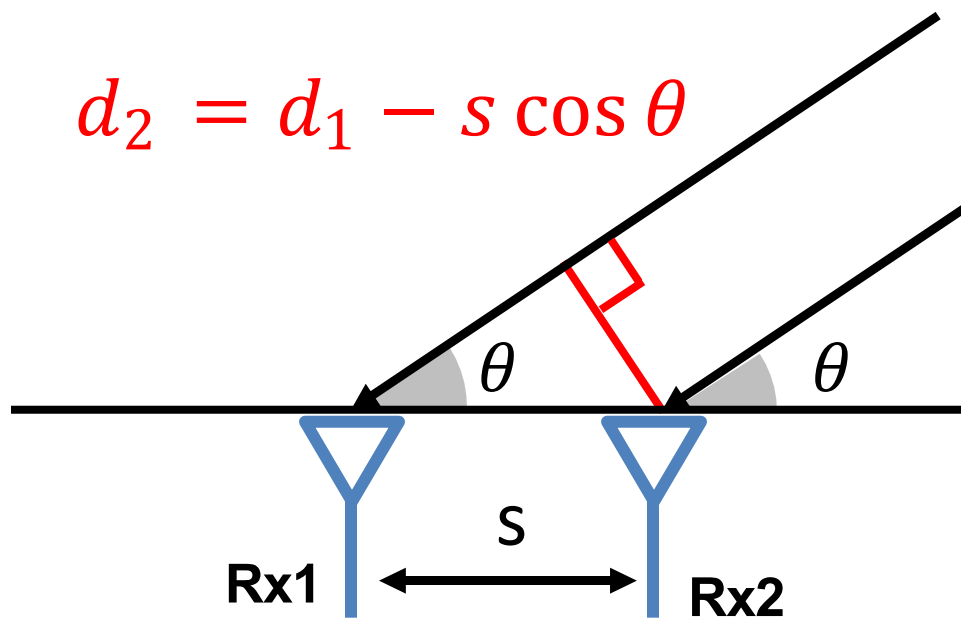
$$\Delta\Phi = \angle h_2 - \angle h_1 = 2\pi s \cos \theta / \lambda \bmod 2\pi$$

Ambiguity: $\exists \theta_1 \neq \theta_2 \mid \Delta\Phi_1 = \Delta\Phi_2 \bmod 2\pi$

To avoid ambiguity, we want: $-\pi \leq \Delta\Phi \leq \pi$

Method 2: AoA Based Localization

Measure Angle of Arrival (AoA) from device to each AP



We want:

$$-\pi \leq \Delta\Phi \leq \pi$$

We have:

$$-1 \leq \cos \theta \leq 1$$

$$-2\pi \frac{s}{\lambda} \leq 2\pi \frac{s}{\lambda} \cos \theta \leq 2\pi \frac{s}{\lambda}$$

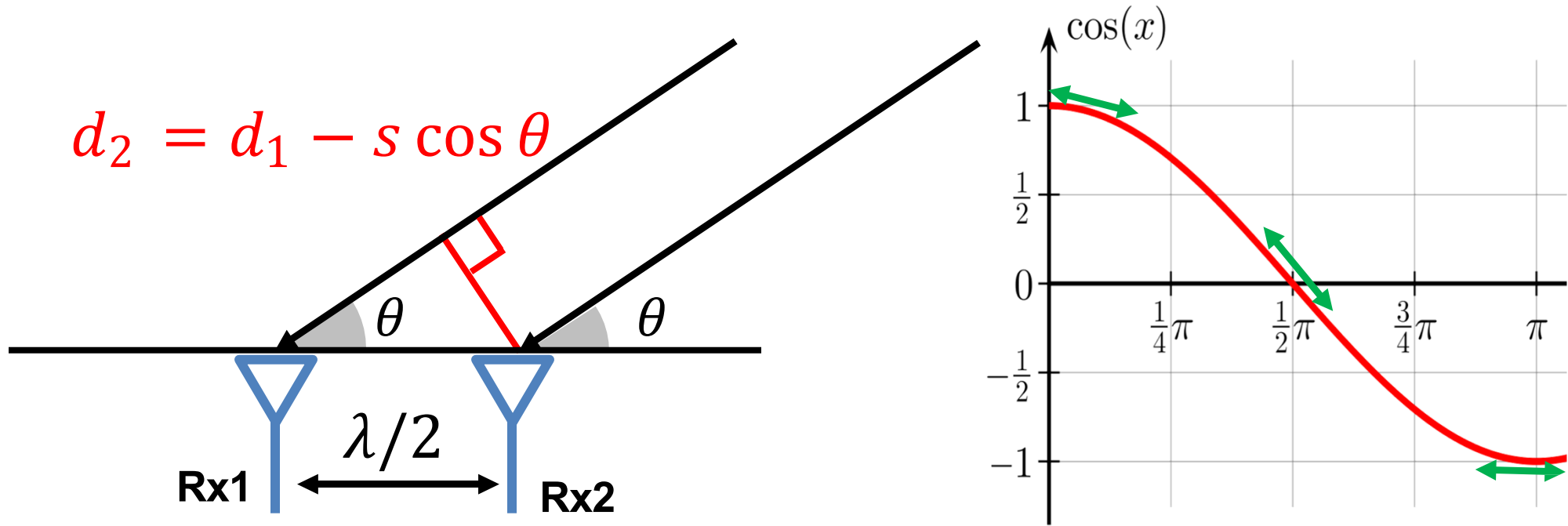
$$-2\pi \frac{s}{\lambda} \leq \Delta\Phi \leq 2\pi \frac{s}{\lambda}$$

$$-\pi \leq \Delta\Phi \leq \pi$$

Set: $s = \lambda/2$

Method 2: AoA Based Localization

Measure Angle of Arrival (AoA) from device to each AP



Pros: More accurate than RSSI, Simple!

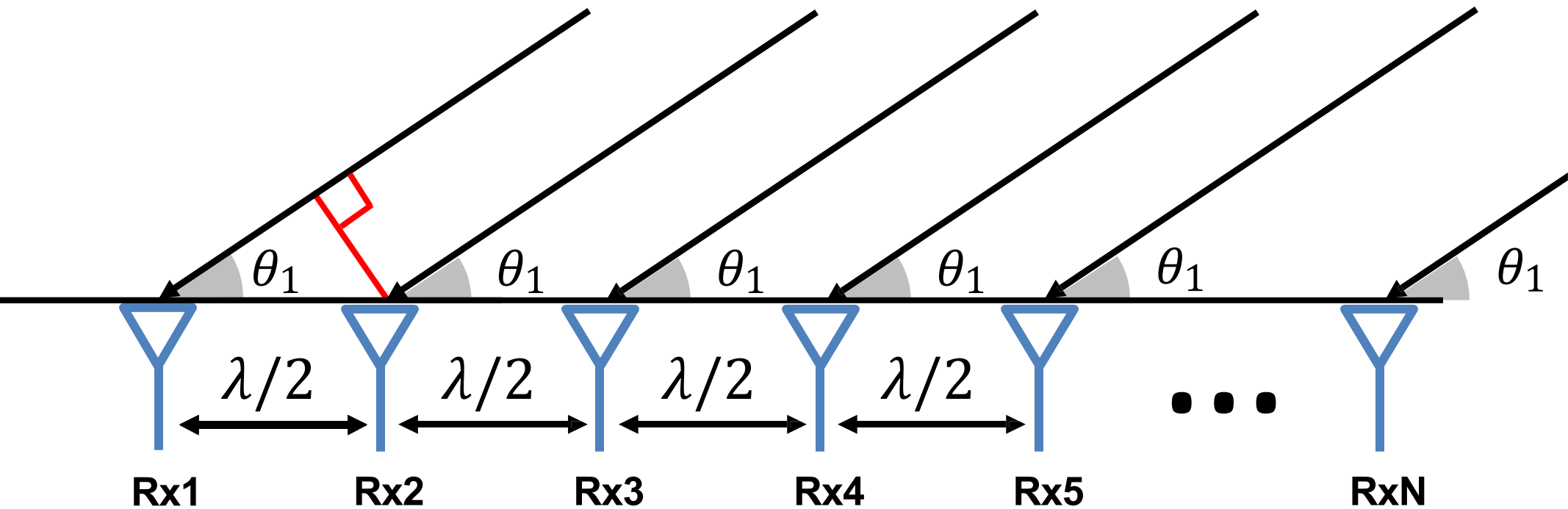
Cons: Ambiguity: $\cos \theta = \cos(-\theta)$

Error not linear with θ due to $\cos \theta$

Requires 2 Antennas separated $\lambda/2$

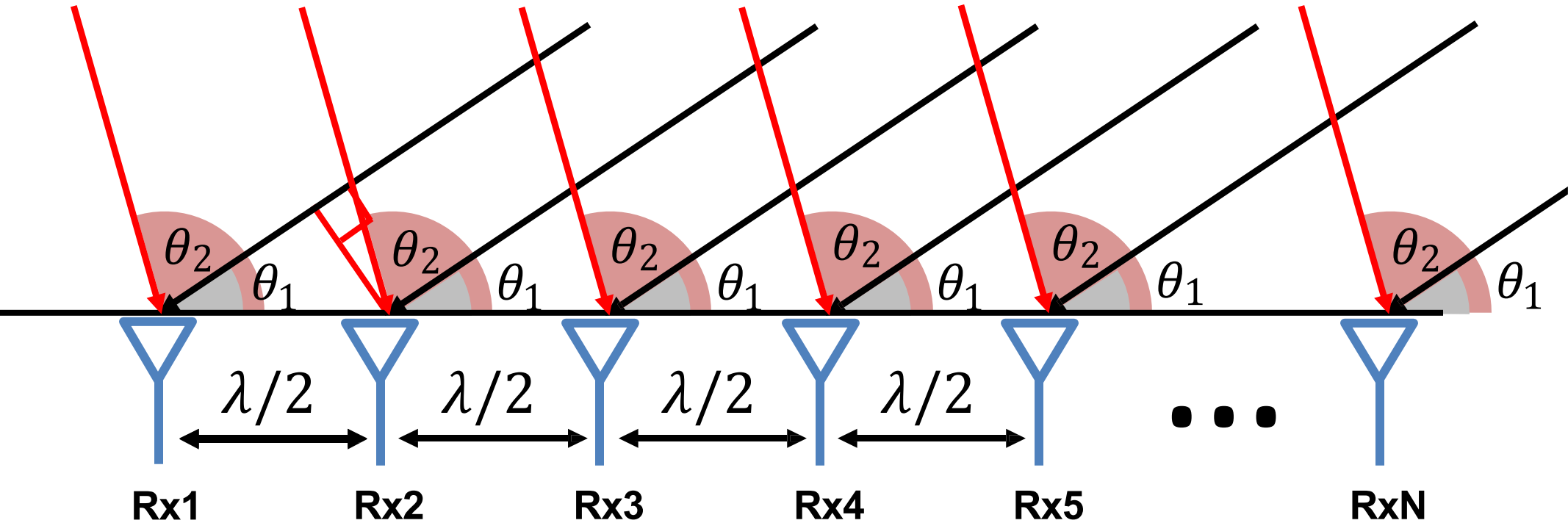
Does not work with multipath!

Method 3: Antenna Arrays (ArrayTrack Paper)



$$h_k = \alpha_1 e^{-j2\pi \frac{d_1 - k s \cos \theta_1}{\lambda}} = \alpha_1 e^{-j\phi_1 + k\pi \cos \theta_1}$$

Method 3: Antenna Arrays (ArrayTrack Paper)



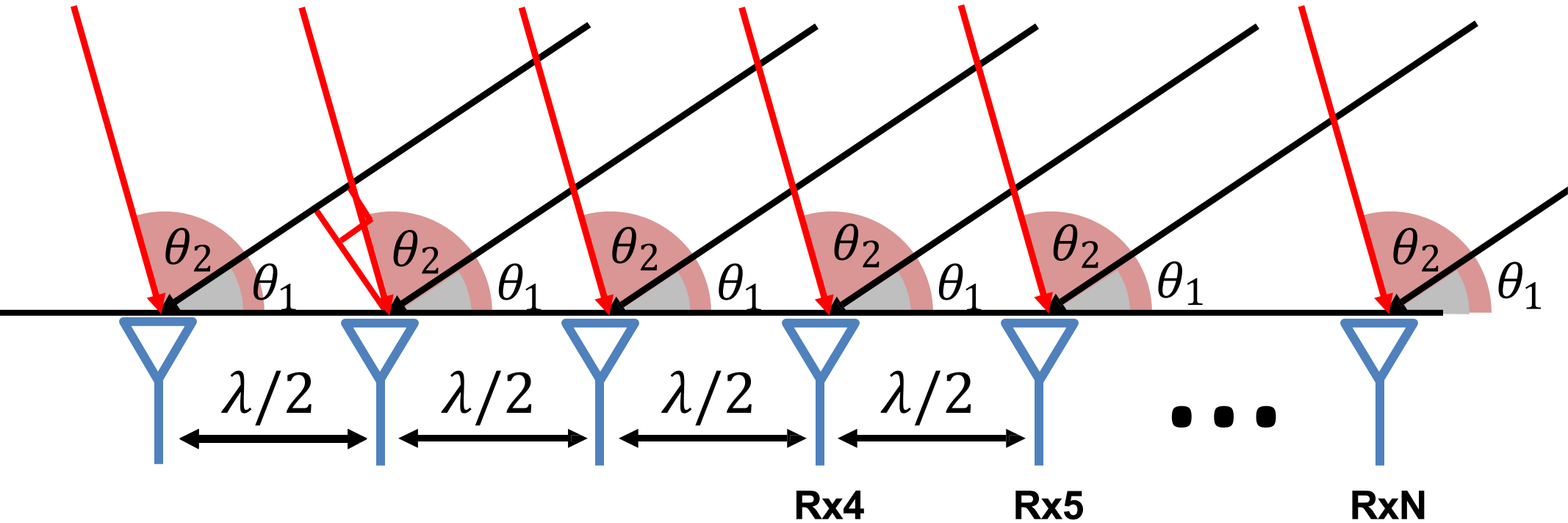
$$h_k = \alpha_1 e^{-j2\pi \frac{d_1 - k s \cos \theta_1}{\lambda}} = \alpha_1 e^{-j\phi_1 + k\pi \cos \theta_1}$$

$$+ \alpha_2 e^{-j2\pi \frac{d_2 - k s \cos \theta_2}{\lambda}} + \alpha_2 e^{-j\phi_2 + k\pi \cos \theta_2}$$

For L paths $\rightarrow$

$$h_k = \sum_{l=1}^L \alpha_l e^{jk\pi \cos \theta_l - j\phi_l}$$

Method 3: Antenna Arrays (ArrayTrack Paper)



$$h_k = \sum_{l=1}^L \alpha_l e^{jk\pi \cos \theta_l - j\phi_l}$$

Multipath Profile: $P(\theta) = \left| \sum_{k=1}^N h_k e^{-jk\pi \cos \theta} \right|^2$

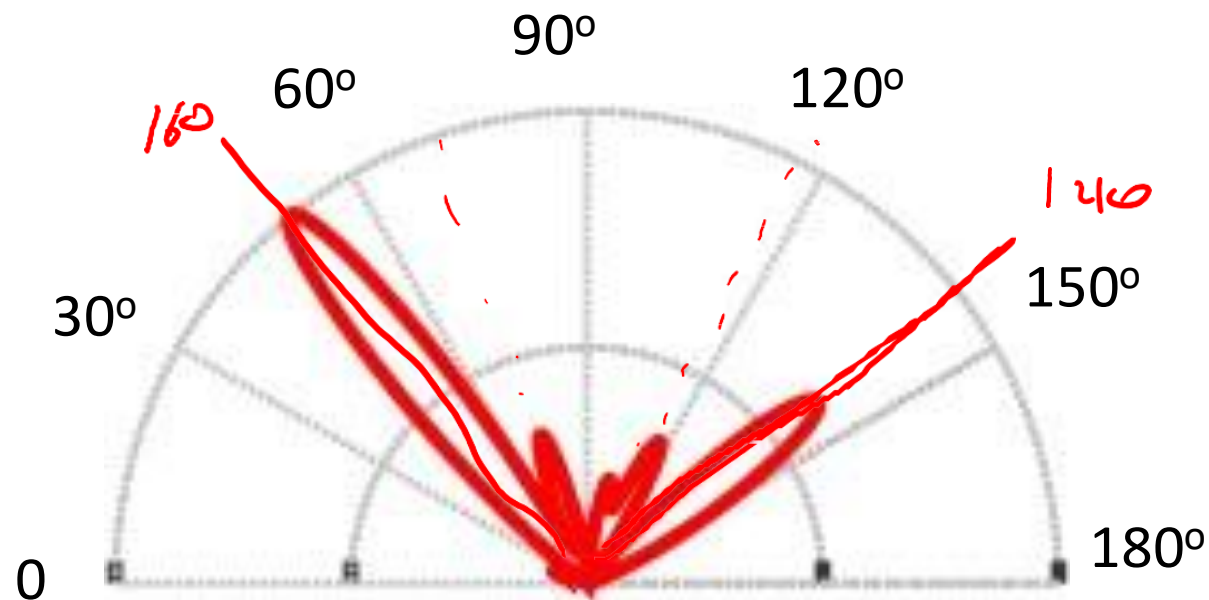
Method 3: Antenna Arrays (ArrayTrack Paper)

$$\begin{aligned}
 P(\theta) &= \left| \sum_{k=1}^N h_k e^{-jk\pi \cos \theta} \right|^2 & h_k &= \sum_{l=1}^L \alpha_l e^{jk\pi \cos \theta_l - j\phi_l} \\
 P(\theta_1) &= \left| \sum_{k=1}^N h_k e^{-jk\pi \cos \theta_1} \right|^2 \\
 &= \left| \sum_{k=1}^N \left(\sum_{l=1}^L \alpha_l e^{jk\pi \cos \theta_l - j\phi_l} \right) e^{-jk\pi \cos \theta_1} \right|^2 \\
 &= \left| \sum_{k=1}^N \alpha_1 e^{-j\phi_1} + \sum_{k=1}^N \left(\sum_{l=2}^L \alpha_l e^{jk\pi (\cos \theta_l - \cos \theta_1) - j\phi_l} \right) \right|^2 \\
 &= \left| \underbrace{N\alpha_1 e^{-j\phi_1}}_{\downarrow} + \sum_{l=2}^L \alpha_l e^{-j\phi_l} \left(\sum_{k=1}^N e^{jk\pi (\cos \theta_l - \cos \theta_1)} \right) \right|^2 \approx N^2 \alpha_1^2
 \end{aligned}$$

Method 3: Antenna Arrays (ArrayTrack Paper)

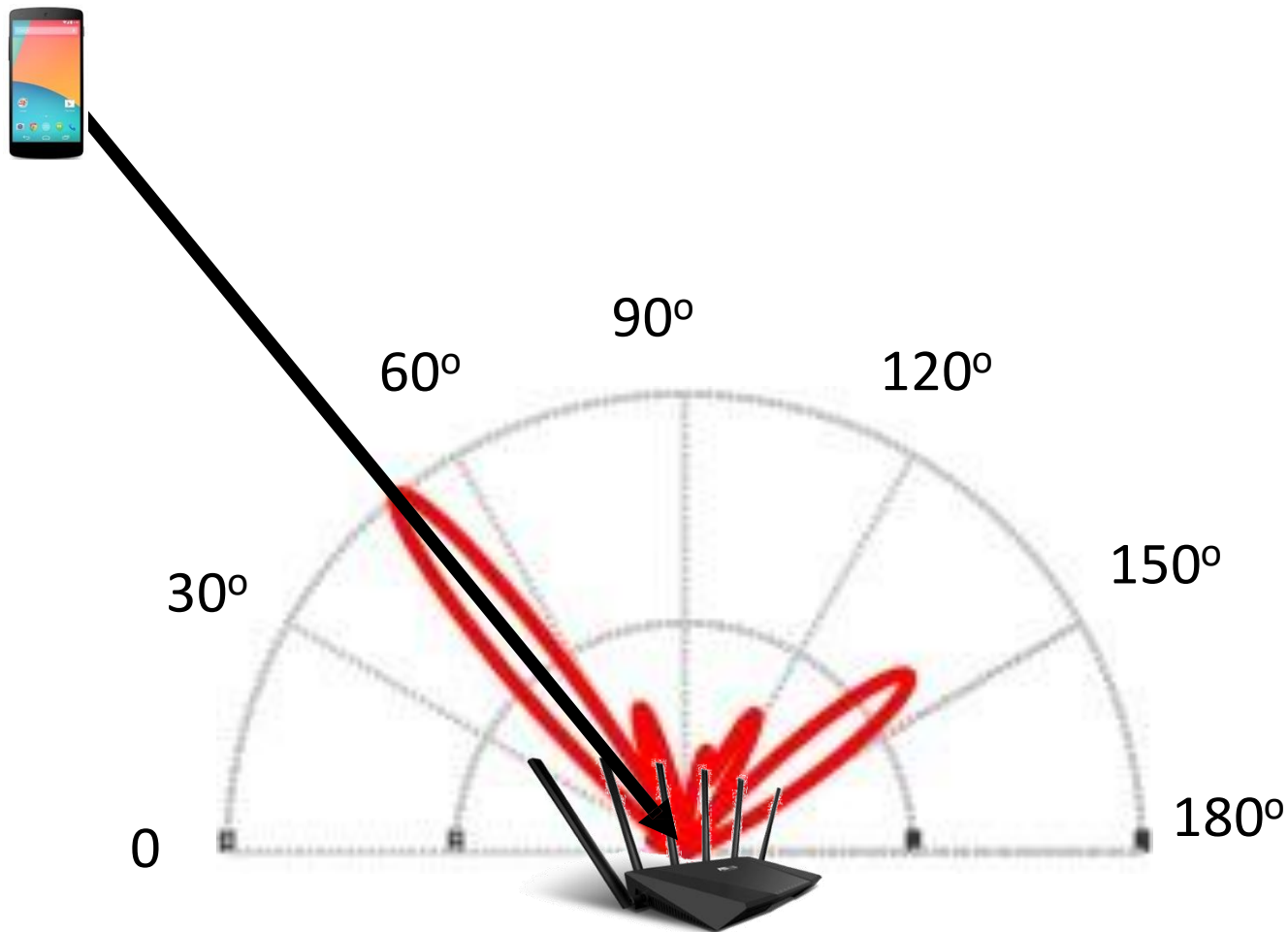
$$h_k = \sum_{l=1}^L \alpha_l e^{jk\pi \cos \theta_l - j\phi_l}$$

$$\text{Multipath Profile: } P(\theta) = \left| \sum_{k=1}^N h_k e^{-jk\pi \cos \theta} \right|^2$$



Method 3: Antenna Arrays (ArrayTrack Paper)

Which is the Line-of-Sight Path (Direct Path)?
Strongest Path!

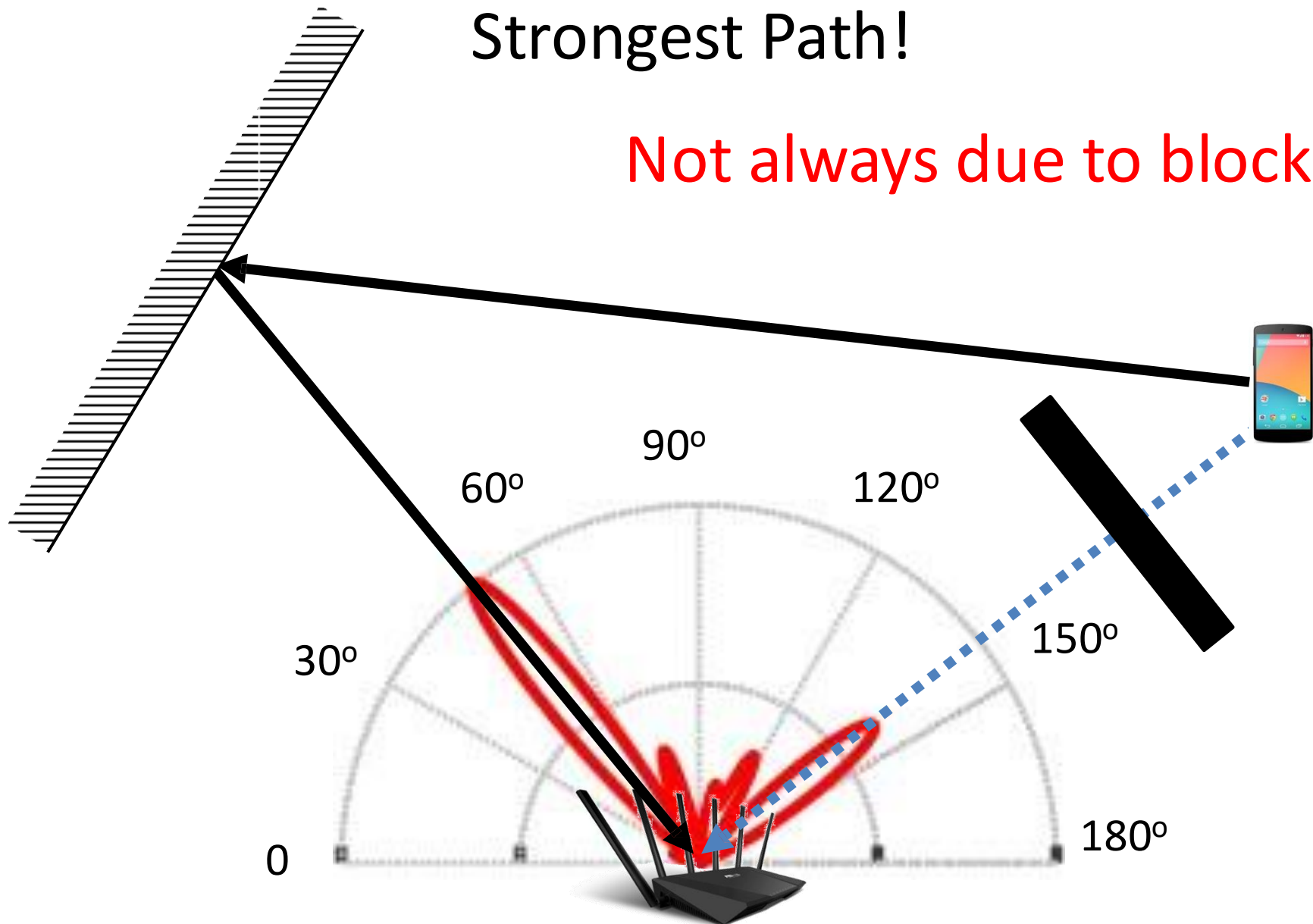


Method 3: Antenna Arrays (ArrayTrack Paper)

Which is the Line-of-Sight Path (Direct Path)?

Strongest Path!

Not always due to blockage!

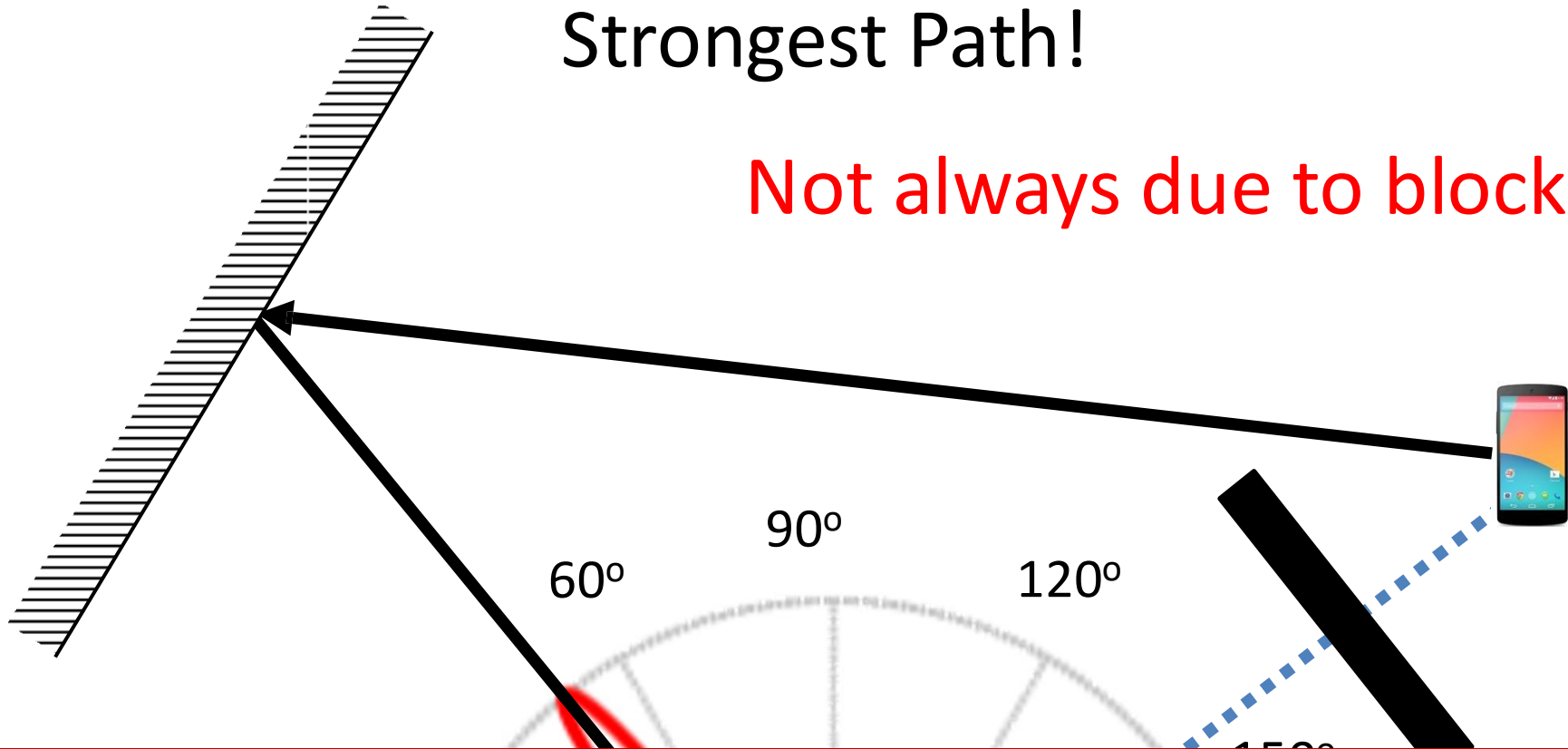


Method 3: Antenna Arrays (ArrayTrack Paper)

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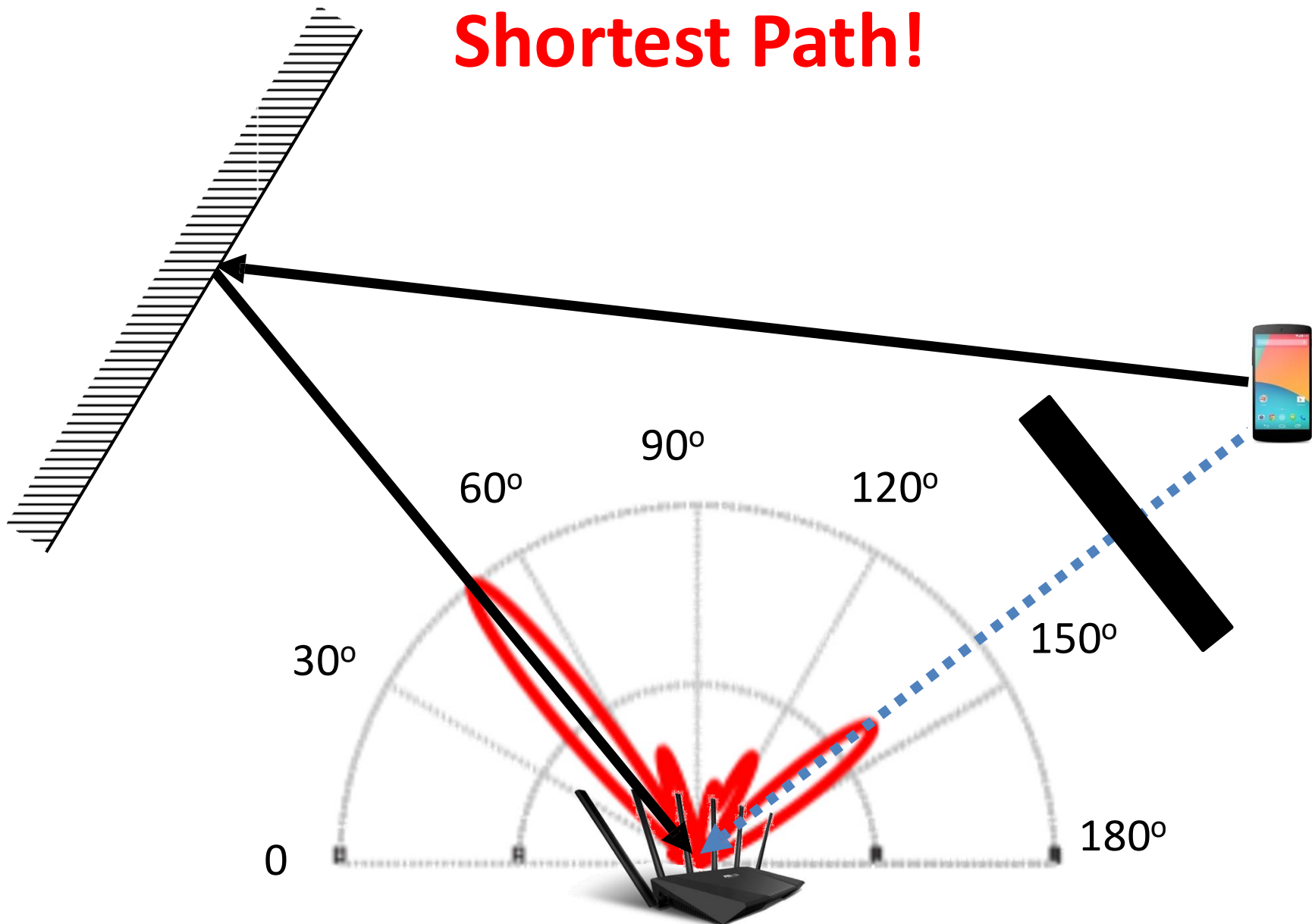
ArrayTrack: Leverage Mobility

- Line of sight path relatively stable with mobility
- Multipath reflection changes faster with mobility

Method 3: Antenna Arrays

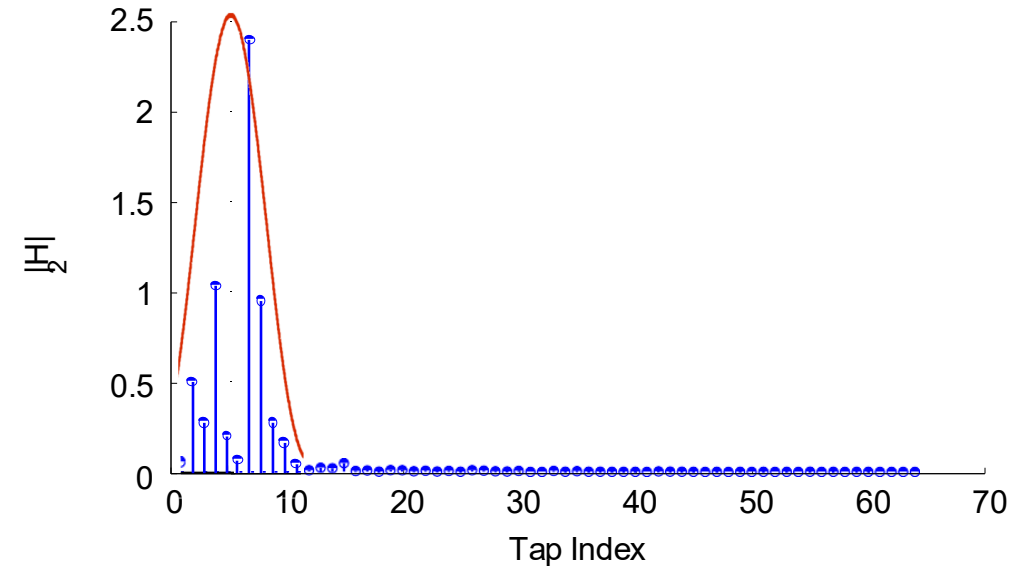
Which is the Line-of-Sight Path (Direct Path)?

Shortest Path!



Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?



Multipath Profile vs Time

$$\Delta r = \frac{\Delta d}{c}$$

$$\Delta d = 1m$$

$$c = 3 \times 10^8 m/s$$

$$\Delta r = 3.3ns$$



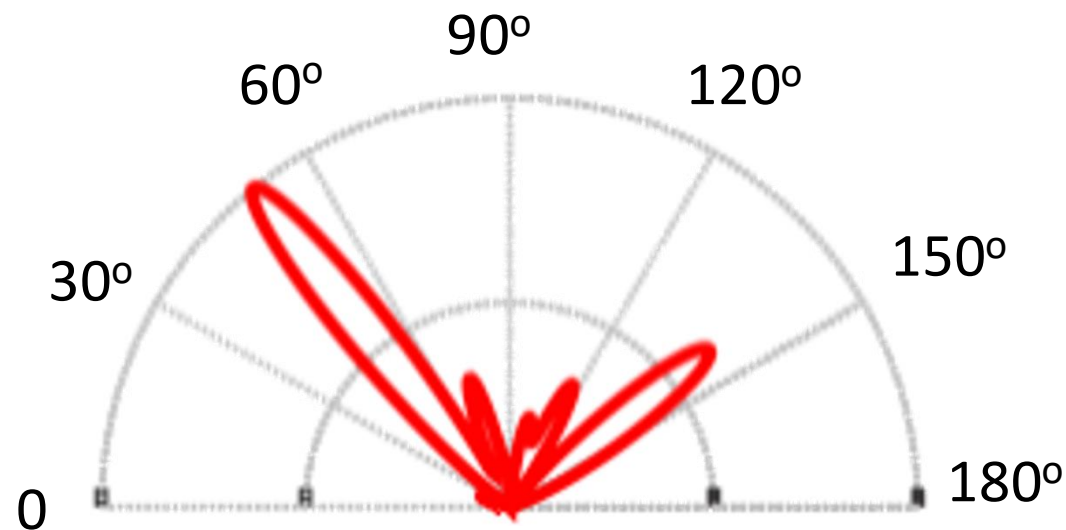
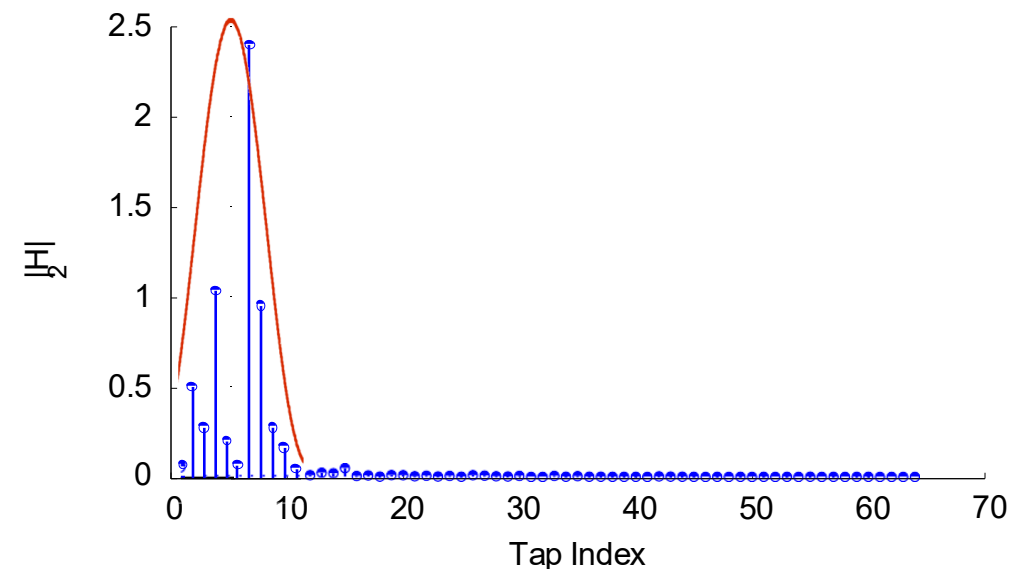
Requires a sampling rate

$$1/\Delta r = 300MHz$$

802.11n bandwidth = 40MHz

Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?



Multipath Profile vs Time

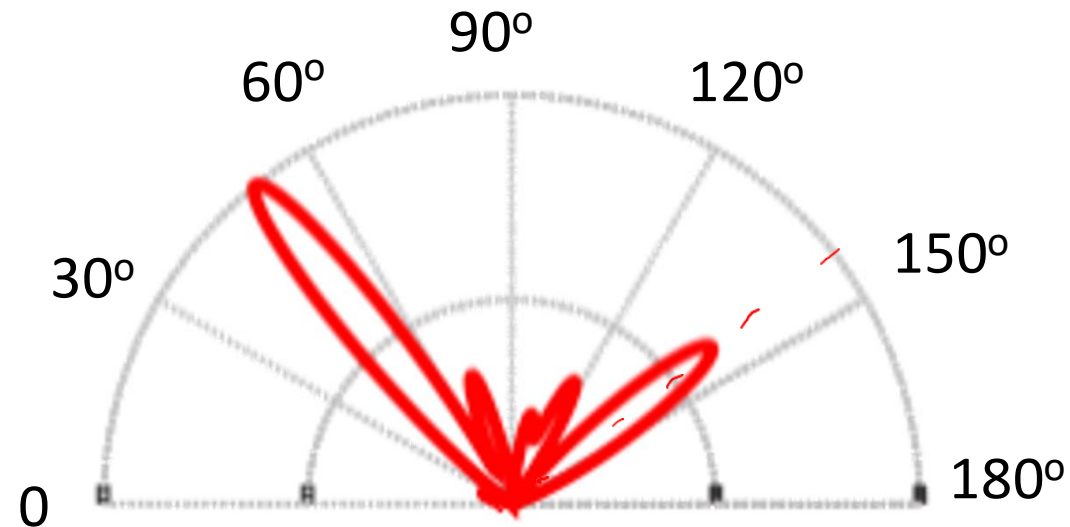
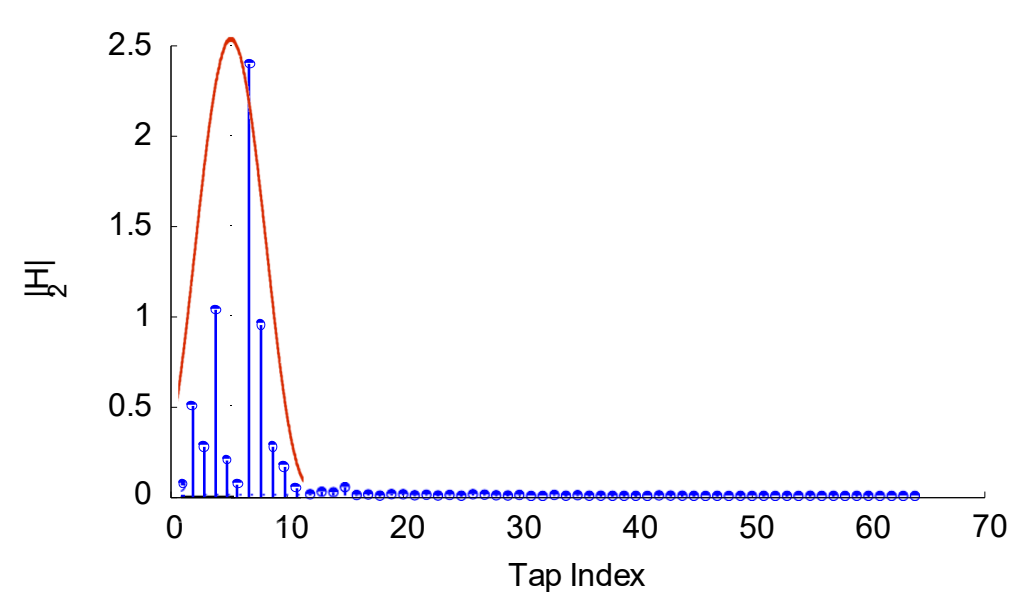
Multipath Profile vs AoA

1. Use multipath profile as a filter to separate different paths
2. Estimate time of arrival of each path
3. Find the shortest path

But How?

Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?



Multipath Profile vs Time

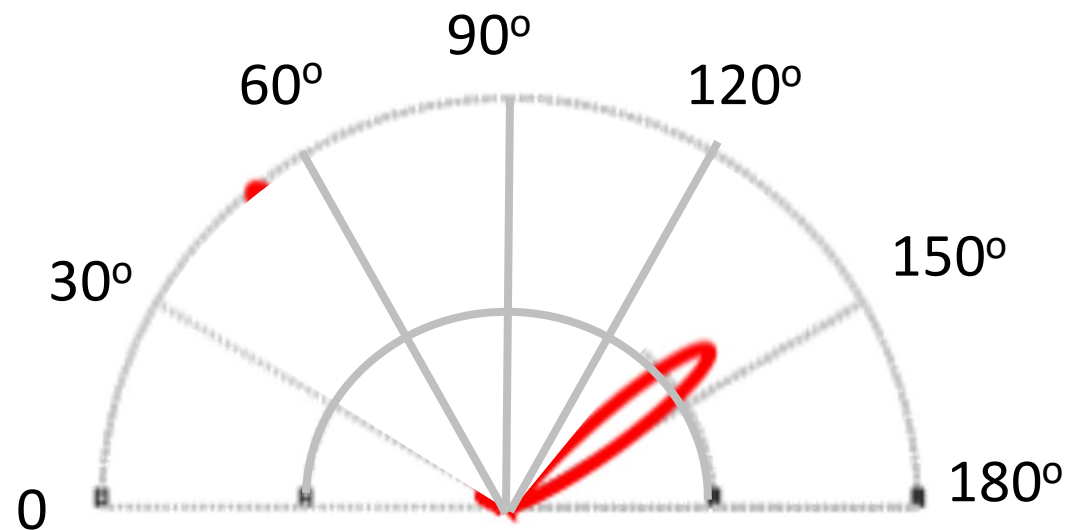
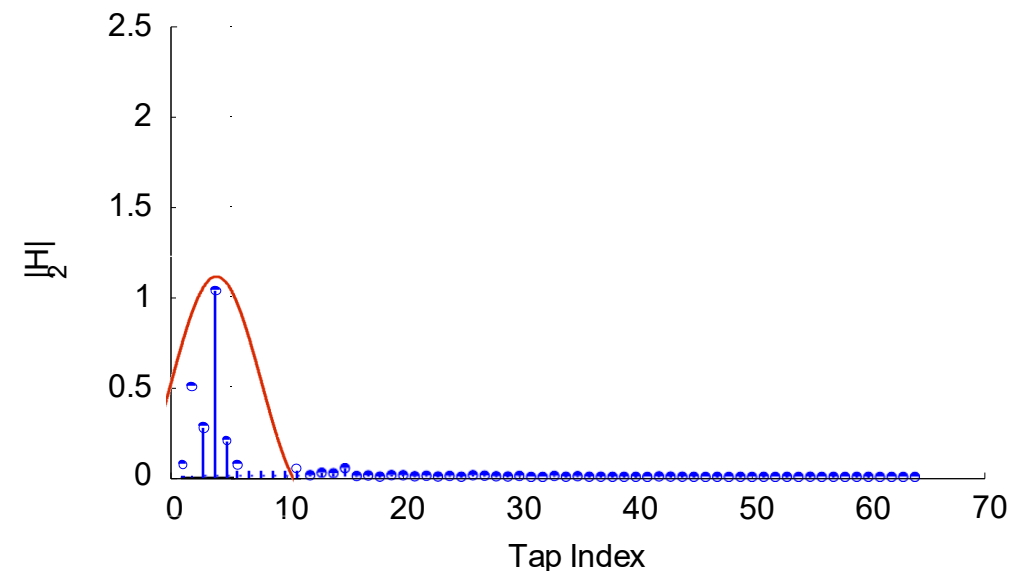
Multipath Profile vs AoA

1. Use multipath profile as a filter to separate different paths

$$y_{\theta_1}(t) = \sum_{k=1}^N y(t) e^{-jk\pi \cos \theta_1}$$

Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?



Multipath Profile vs Time

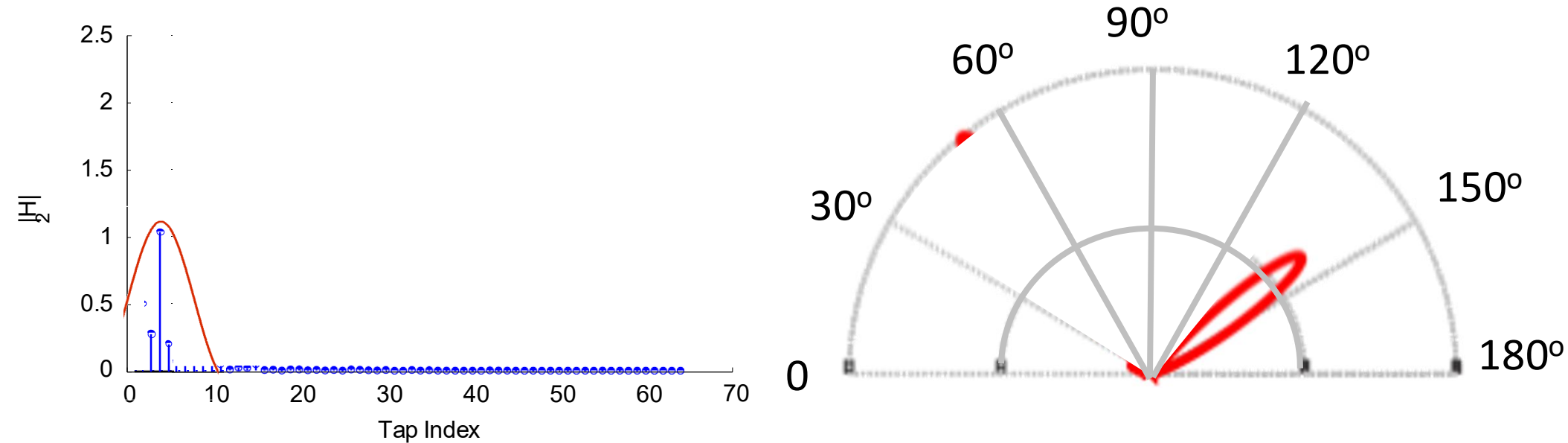
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Multipath Profile vs Time

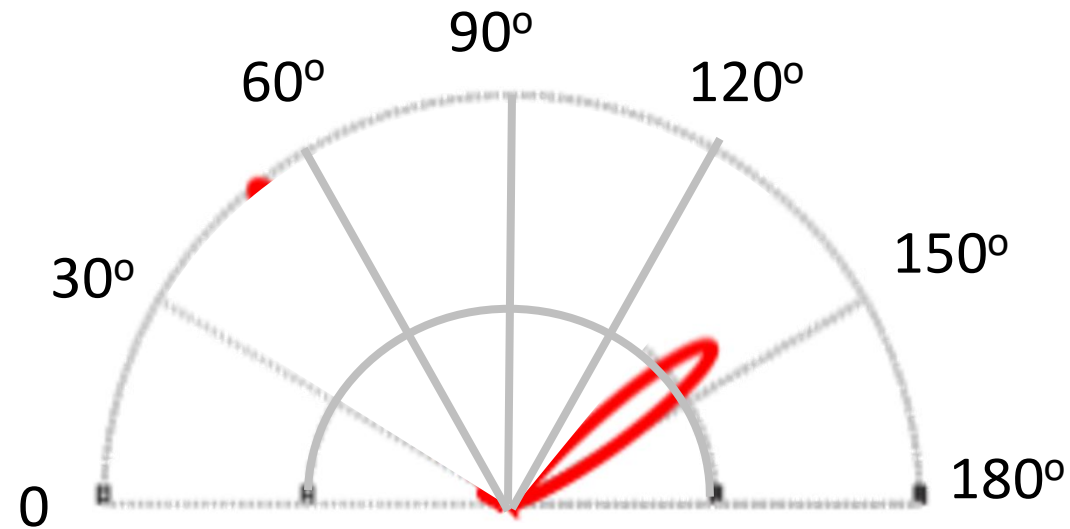
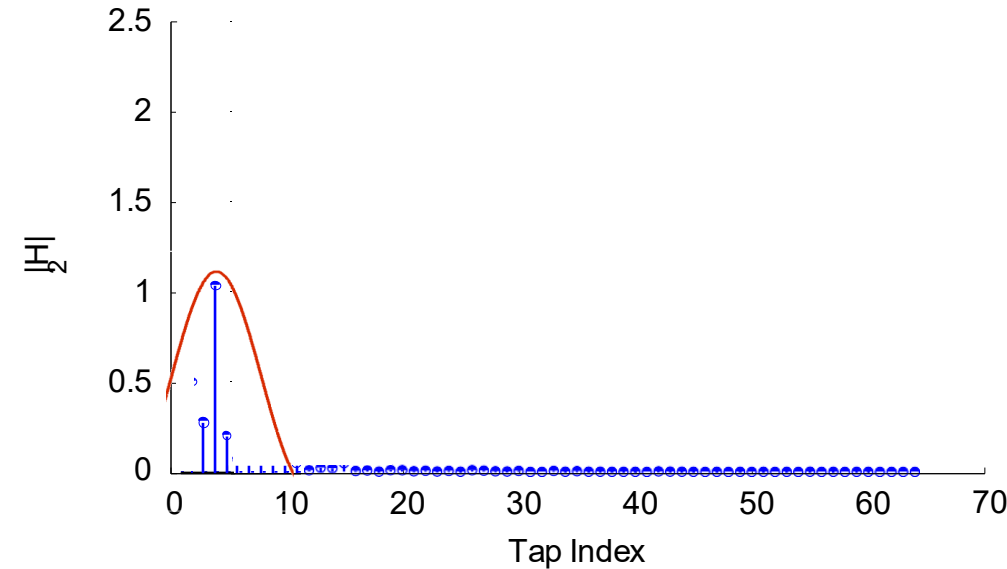
Multipath Profile vs AoA

1. Use multipath profile as a filter to separate different paths
2. Estimate time of arrival of each path

Time Resolution still not enough ➡ Use OFDM

Method 3: Antenna Arrays

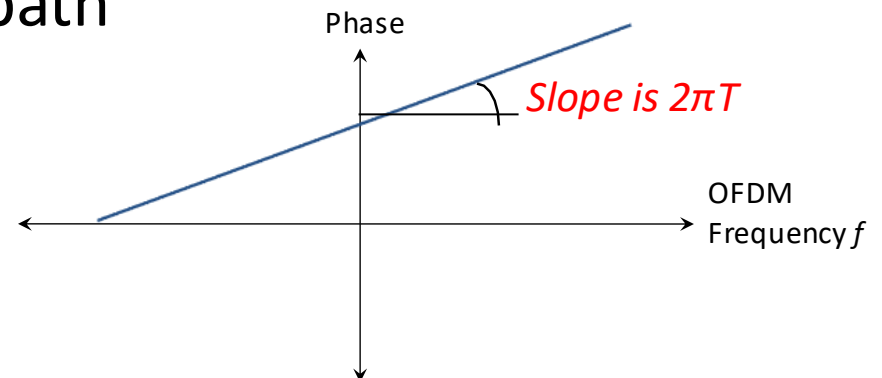
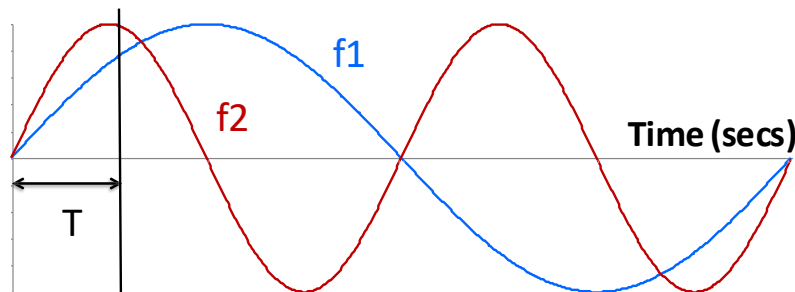
Which is the Shortest Path (Direct Path)?



Multipath Profile vs Time

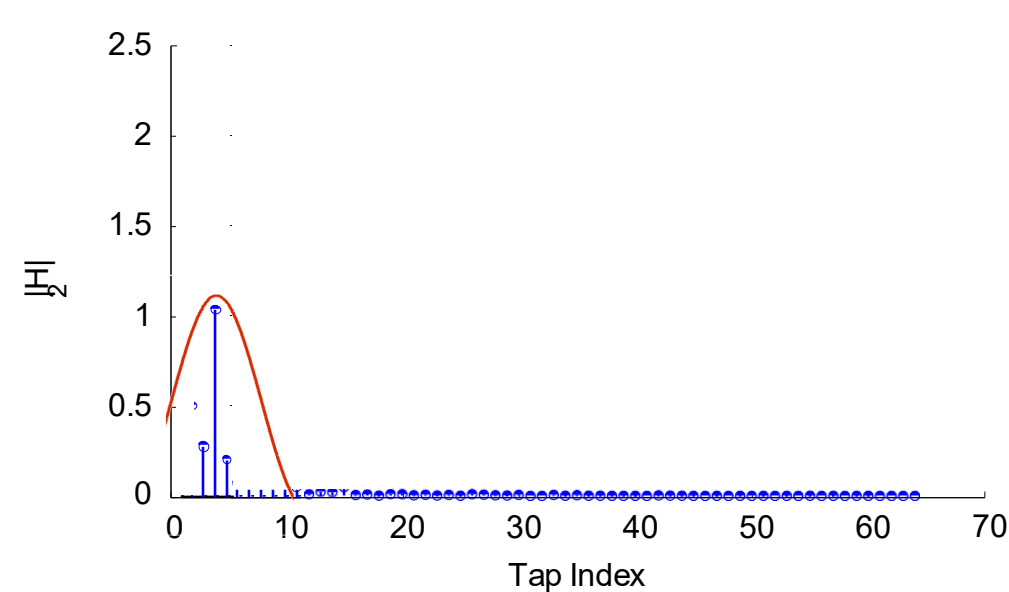
Multipath Profile vs AoA

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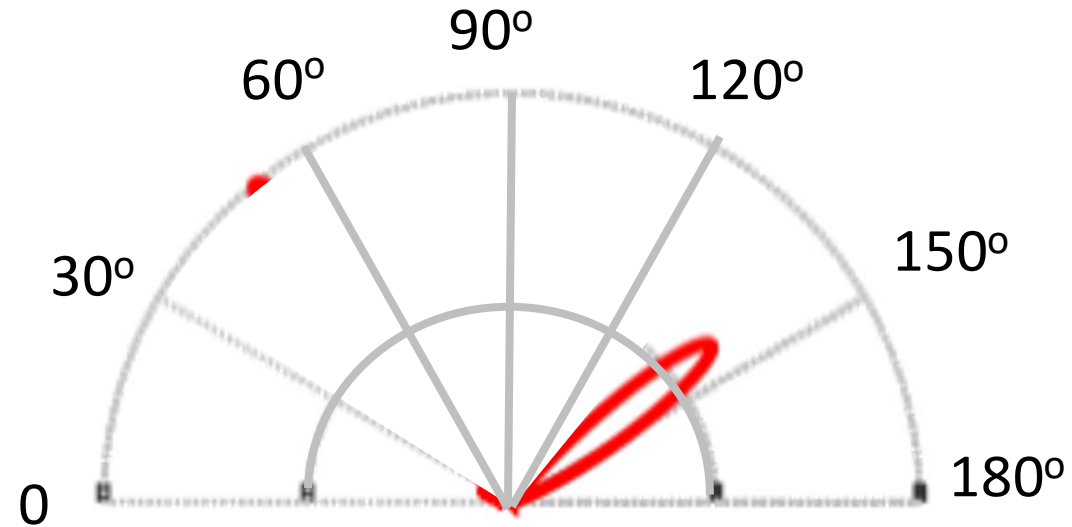


Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?



Multipath Profile vs Time



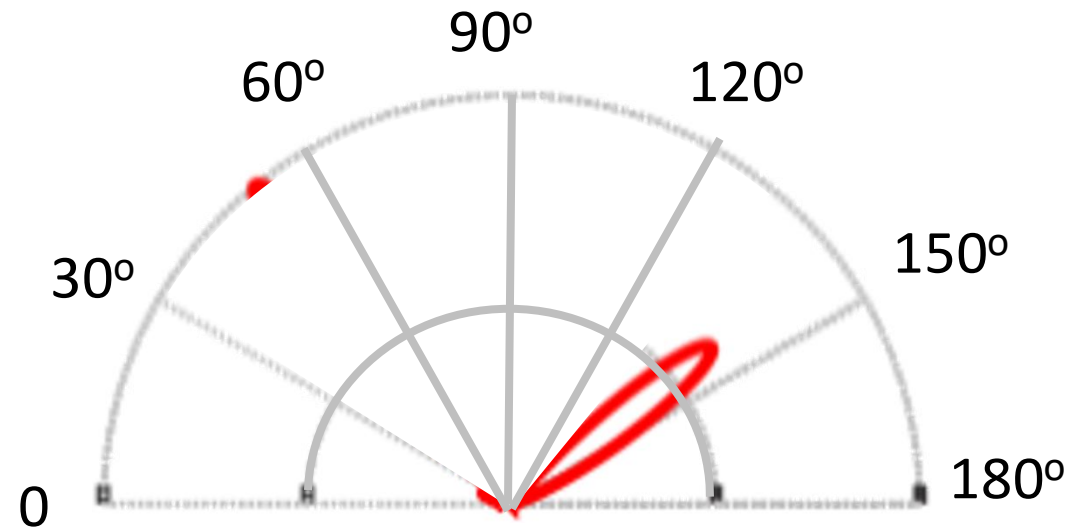
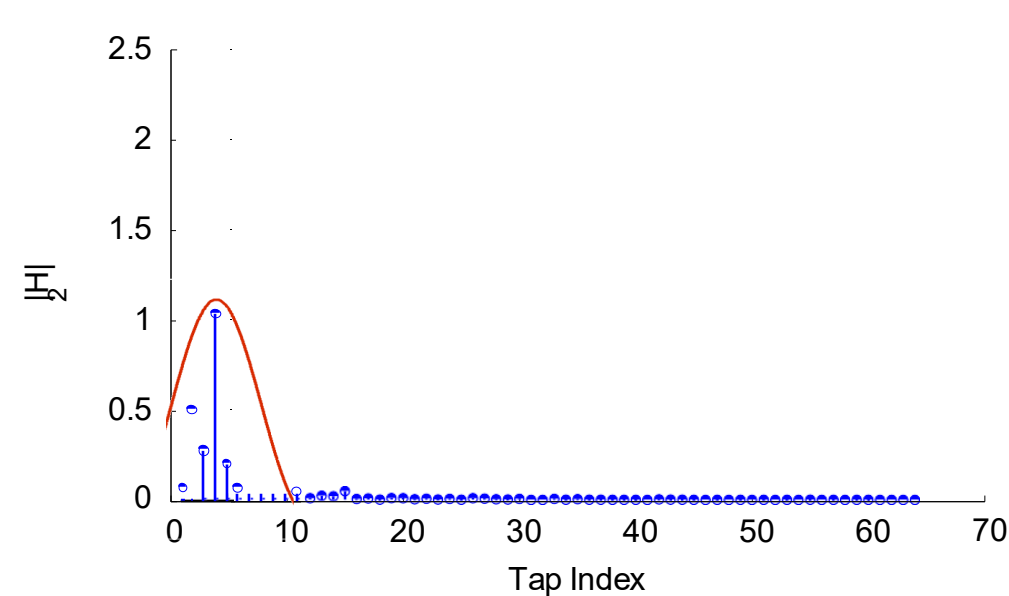
Multipath Profile vs AoA

1. Use multipath profile as a filter to separate different paths
2. Estimate time of arrival of each path
 - Use OFDM to estimate delay from slope of phase vs freq.

But this delay includes packet detection delay & processing delay, not just propagation delay!

Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?

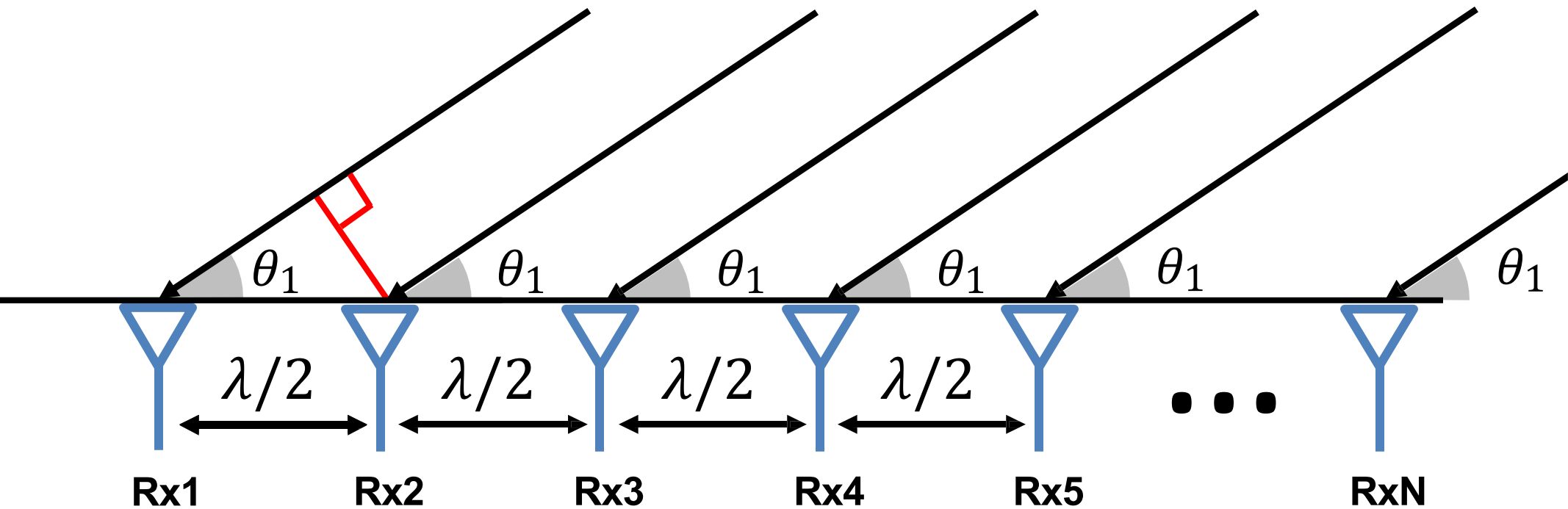


Multipath Profile vs Time

Multipath Profile vs AoA

1. Use multipath profile as a filter to separate different paths
2. Estimate time of arrival of each path
 - Use OFDM to estimate delay from slope of phase vs freq.
 - Compute relative delay for different paths!
3. Compare relative delays to find the shortest path

Method 3: Antenna Arrays



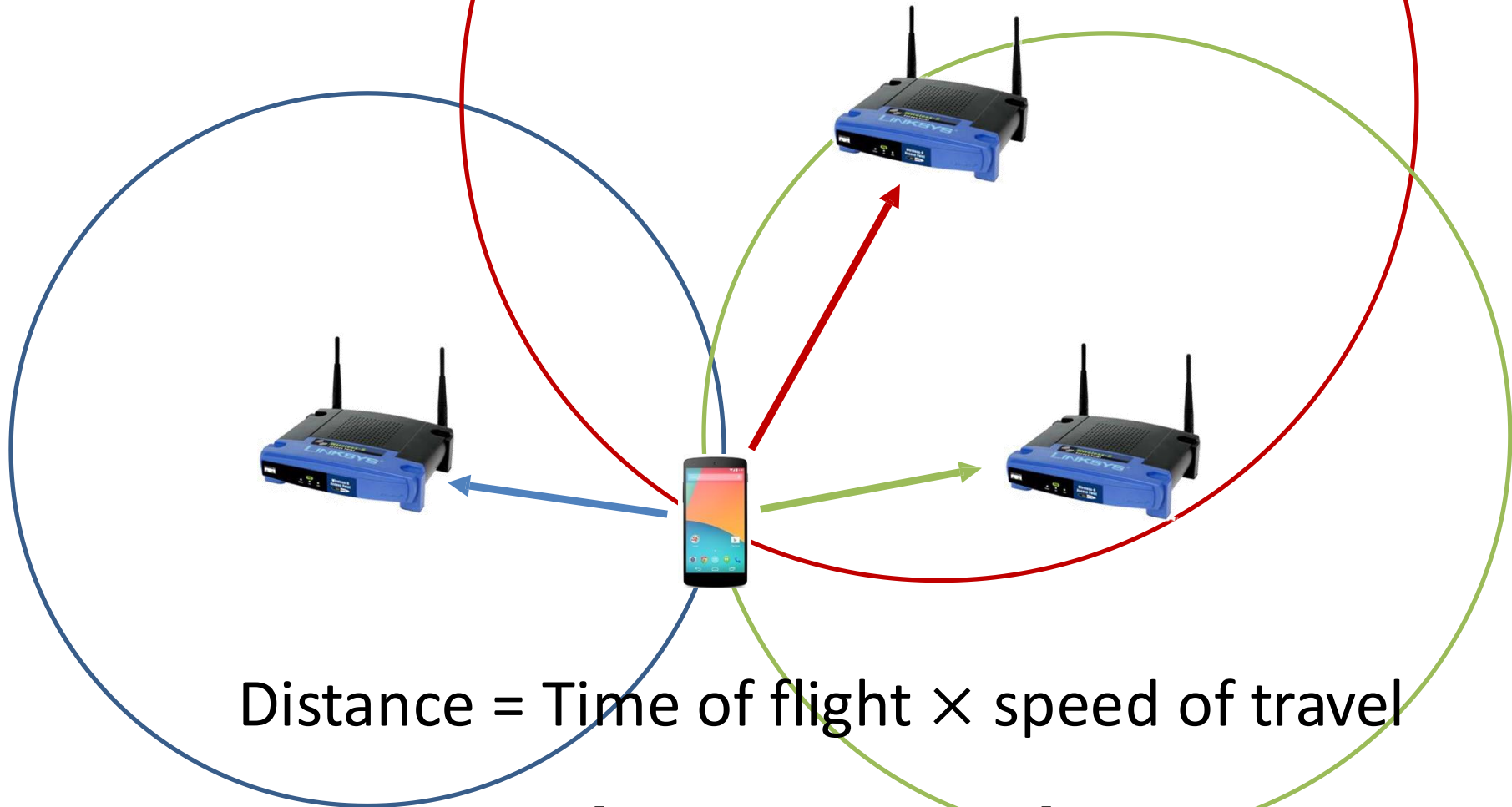
Pros: Works with multipath, No need for fingerprinting

Cons: Requires more hardware!

Assumes device is sufficiently far such that wavefront is parallel

Method 4: ToF Based Localization

Measure Time of Flight (ToF) from device to each AP



Distance = Time of flight $\times$ speed of travel

Measure ToF $\rightarrow$ Get distance $\rightarrow$ Trilateration

Method 4: ToF Based Localization

Measure Time of Flight (ToF) from device to each AP

Challenges:

- How do you know when signal was transmitted?



Measure Round Trip

- How about packet detection delay & processing delay?
 - Use OFDM to correct for packet detection delay
 - Estimate and calibrate for processing delay

Not Practical!

Method 4: ToF Based Localization

Measure Time of Flight (ToF) from device to each AP

Challenges:

- How do you know when signal was transmitted?



Method 4: ToF Based Localization

Measure Time of Flight (ToF) from device to each AP

Challenges:

- How do you know when signal was transmitted?



Method 4: ToF Based Localization

Measure Time of Flight (ToF) from device to each AP

Challenges:

- Accuracy limited by sampling rate (bandwidth)!

$$\Delta d = \Delta \tau \times c$$

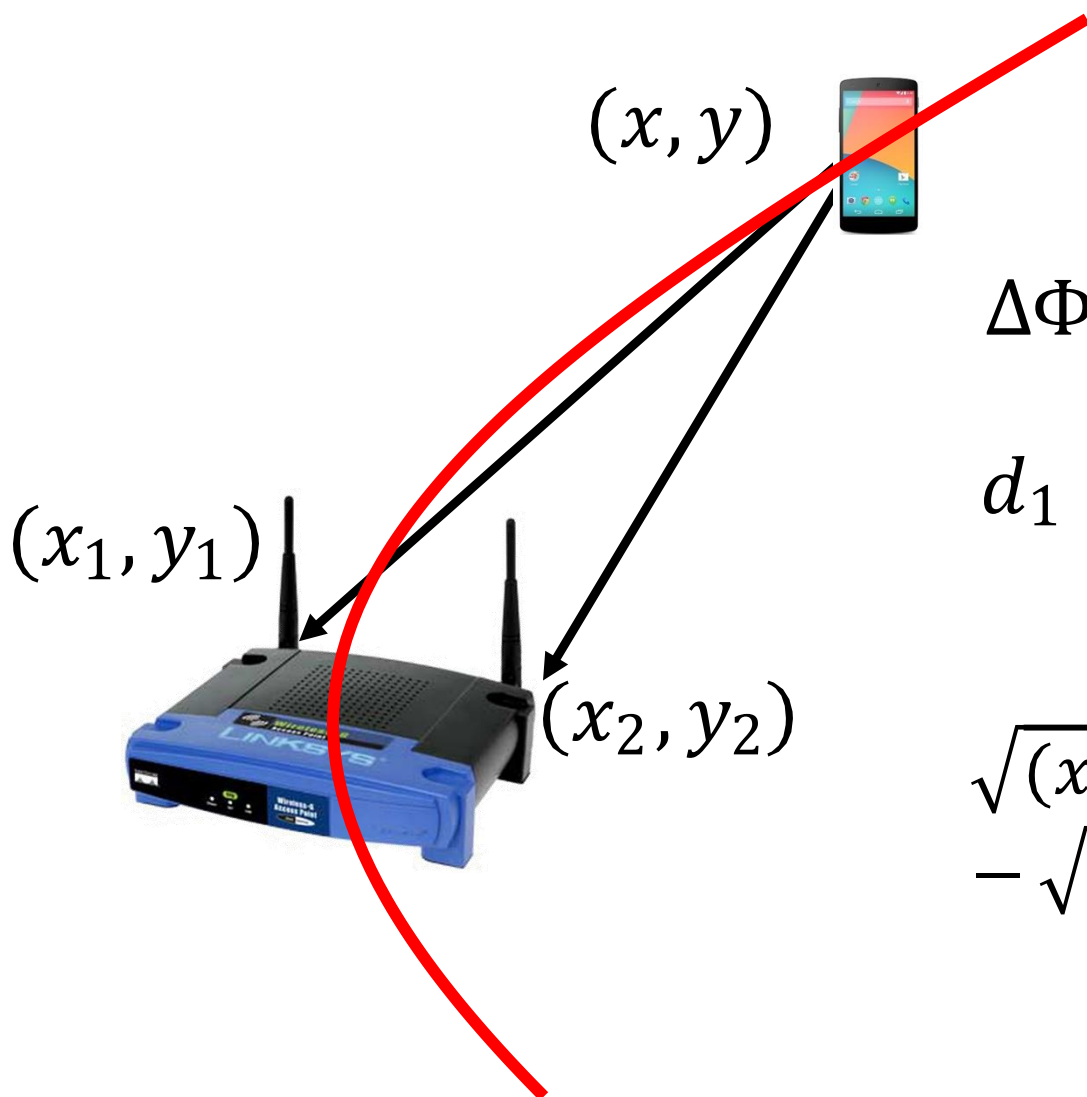
802.11n bandwidth = $40MHz$ $\Rightarrow \Delta \tau = 25ns$ $\Rightarrow \Delta d = 12.5\ m$

- Other systems than WiFi can get accurate ToF:
 - UWB: Ultra-Wide Band
 - FMCW: Frequency Modulated Carrier Wave

Not Supported in WiFi
(Will discuss in future lectures)

Method 5: TDoA Based Localization

Measure Time Difference of Arrival (TDoA) from device to AP's antennas



$$h_1 \propto e^{-j2\pi\frac{d_1}{\lambda}} \quad h_2 \propto e^{-j2\pi\frac{d_2}{\lambda}}$$

$$\Delta\Phi = \angle h_2 - \angle h_1$$

$$d_1 - d_2 = \lambda\Delta\Phi/2\pi = d'$$

Equation of Hyperbola

$$\sqrt{(x - x_1)^2 + (y - y_1)^2} - \sqrt{(x - x_2)^2 + (y - y_2)^2} = d'$$

Method 5: TDoA Based Localization

Measure Time Difference of Arrival (TDoA) from device to AP's antennas

